

(12) **United States Patent**
Saeki et al.

(10) **Patent No.: US 12,416,434 B2**
(45) **Date of Patent: Sep. 16, 2025**

(54) **REFRIGERANT VESSEL AND
REFRIGERATION CYCLE APPARATUS**

(71) Applicant: **DAIKIN INDUSTRIES, LTD.**, Osaka
(JP)

(72) Inventors: **Kumiko Saeki**, Osaka (JP); **Yuta
Iyoshi**, Osaka (JP); **Yoshiki Yamanoi**,
Osaka (JP)

(73) Assignee: **Daikin Industries, Ltd.**, Osaka (JP)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/622,036**

(22) Filed: **Mar. 29, 2024**

(65) **Prior Publication Data**

US 2024/0240843 A1 Jul. 18, 2024

Related U.S. Application Data

(63) Continuation of application No.
PCT/JP2022/035473, filed on Sep. 22, 2022.

(30) **Foreign Application Priority Data**

Sep. 30, 2021 (JP) 2021-161999

(51) **Int. Cl.**
F25B 41/40 (2021.01)
F25B 43/00 (2006.01)

(Continued)

(52) **U.S. Cl.**
CPC **F25B 41/40** (2021.01); **F25B 43/006**
(2013.01); **F25B 49/022** (2013.01); **F25B 7/00**
(2013.01);

(Continued)

(58) **Field of Classification Search**
CPC B01D 45/00; B01D 45/02; B01D 45/12
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,167,128 A * 12/1992 Bottum G01F 23/74
62/220
5,426,956 A * 6/1995 Phillippe F25B 41/00
62/509

(Continued)

FOREIGN PATENT DOCUMENTS

CN 110770516 A 2/2020
CN 111023610 A 4/2020

(Continued)

OTHER PUBLICATIONS

International Search Report for PCT/JP2022/035473 (PCT/ISA/
210) mailed on Nov. 22, 2022.

(Continued)

Primary Examiner — Jerry-Daryl Fletcher

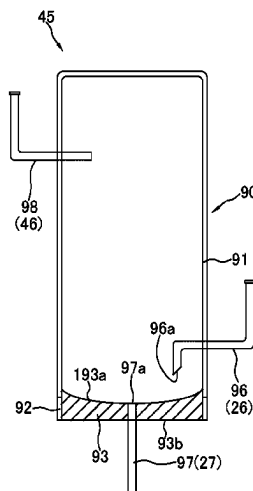
Assistant Examiner — Daniel C Comings

(74) *Attorney, Agent, or Firm* — Birch, Stewart, Kolasch
& Birch, LLP

(57) **ABSTRACT**

Provided are a refrigerant vessel and a refrigeration cycle
apparatus capable of suppressing accumulation of a refrig-
erating machine oil inside. A secondary-side receiver
reserves a refrigerant in a secondary-side refrigerant circuit
in which a refrigerant and a refrigerating machine oil cir-
culate, the secondary-side receiver including a vessel body,
a first refrigerant pipe connected to the vessel body, and a
second refrigerant pipe connected to the vessel body, in
which the refrigerating machine oil is incompatible with the
refrigerant and has a higher density than the refrigerant, and
the second refrigerant pipe extends downward from a bot-
tom part of the vessel body.

20 Claims, 13 Drawing Sheets



(51) **Int. Cl.**

F25B 49/02 (2006.01)
F25B 7/00 (2006.01)
F25B 9/00 (2006.01)

(52) **U.S. Cl.**

CPC *F25B 9/008* (2013.01); *F25B 2400/02*
 (2013.01); *F25B 2400/16* (2013.01); *F25B*
2400/23 (2013.01); *F25B 2600/25* (2013.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,457,325 B1 10/2002 Vetter
 8,505,331 B2 * 8/2013 Pham F04C 29/042
 62/509
 9,415,335 B2 * 8/2016 Huff F25B 1/10
 2005/0115270 A1 6/2005 Furuta et al.
 2012/0117996 A1 * 5/2012 Hinde F25B 7/00
 62/468
 2014/0360223 A1 * 12/2014 Setoguchi F25B 43/00
 62/509
 2018/0017298 A1 * 1/2018 Dziubinski B01D 15/26
 2019/0078795 A1 3/2019 Kim et al.
 2020/0271357 A1 * 8/2020 Miyawaki F25B 13/00
 2021/0333021 A1 10/2021 Matsuoka et al.

FOREIGN PATENT DOCUMENTS

EP 2631562 A1 * 8/2013 F24D 19/1039
 JP 2000-205704 A 7/2000
 JP 2000-241049 A 9/2000
 JP 2005-153773 A 6/2005
 JP 2008-164225 A 7/2008
 JP 2008-286505 A 11/2008
 JP 2009-276018 A 11/2009
 JP 2013-148328 A 8/2013
 JP 2015-224832 A 12/2015
 JP 2017020768 A * 1/2017 F25B 43/02
 JP 2018-112367 A 7/2018
 JP 2020-85424 A 6/2020
 JP 2020-94698 A 6/2020
 WO WO-2016113785 A1 * 7/2016 F04C 18/356
 WO WO-2018173255 A1 * 9/2018 F25B 13/00

OTHER PUBLICATIONS

International Preliminary Report on Patentability and English translation of the Written Opinion of the International Searching Authority for International Application No. PCT/JP2022/035473, dated Apr. 11, 2024.

Extended European Search Report for European Application No. 22876058.3, dated Jan. 21, 2025.

* cited by examiner

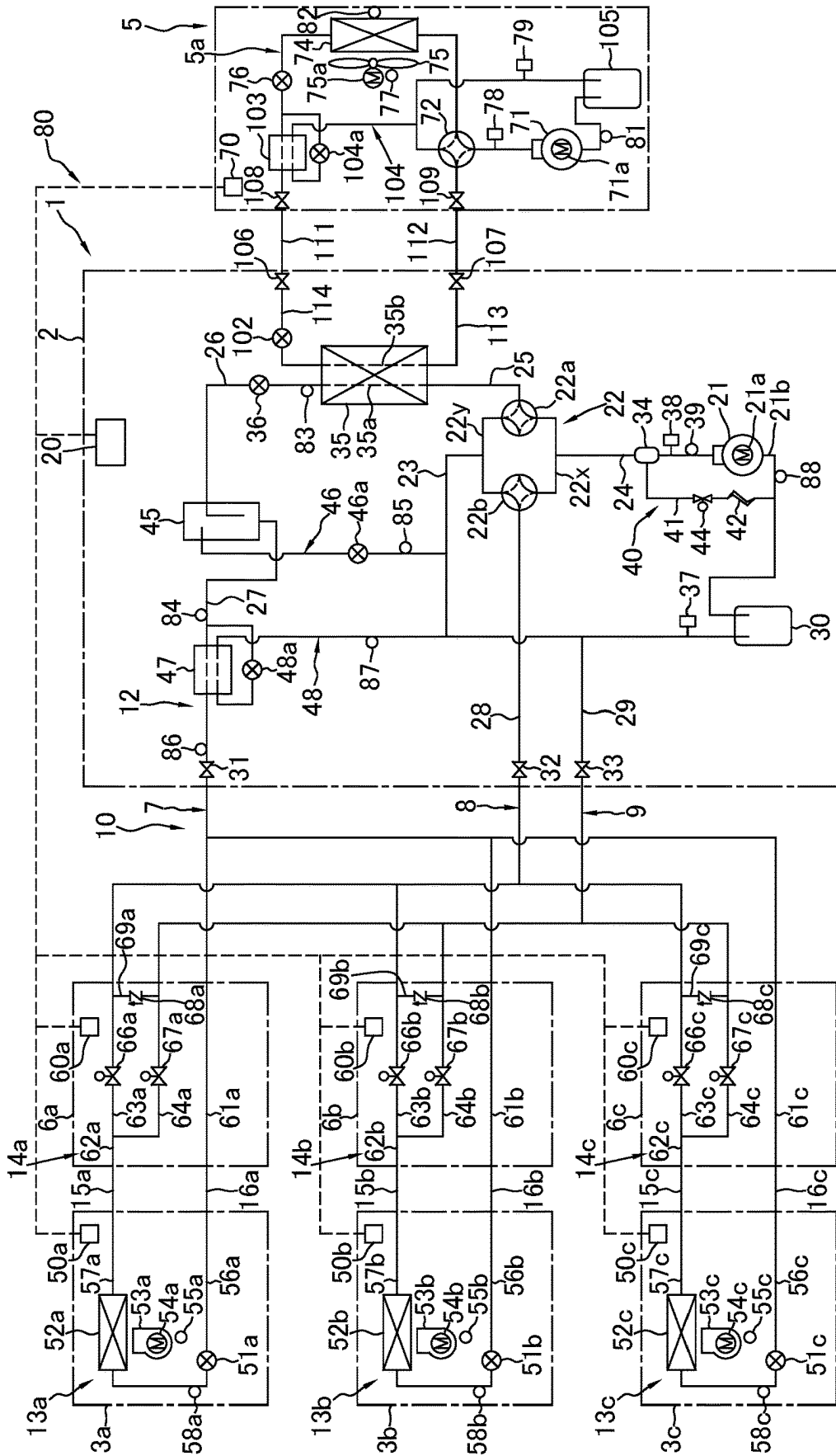


FIG. 1

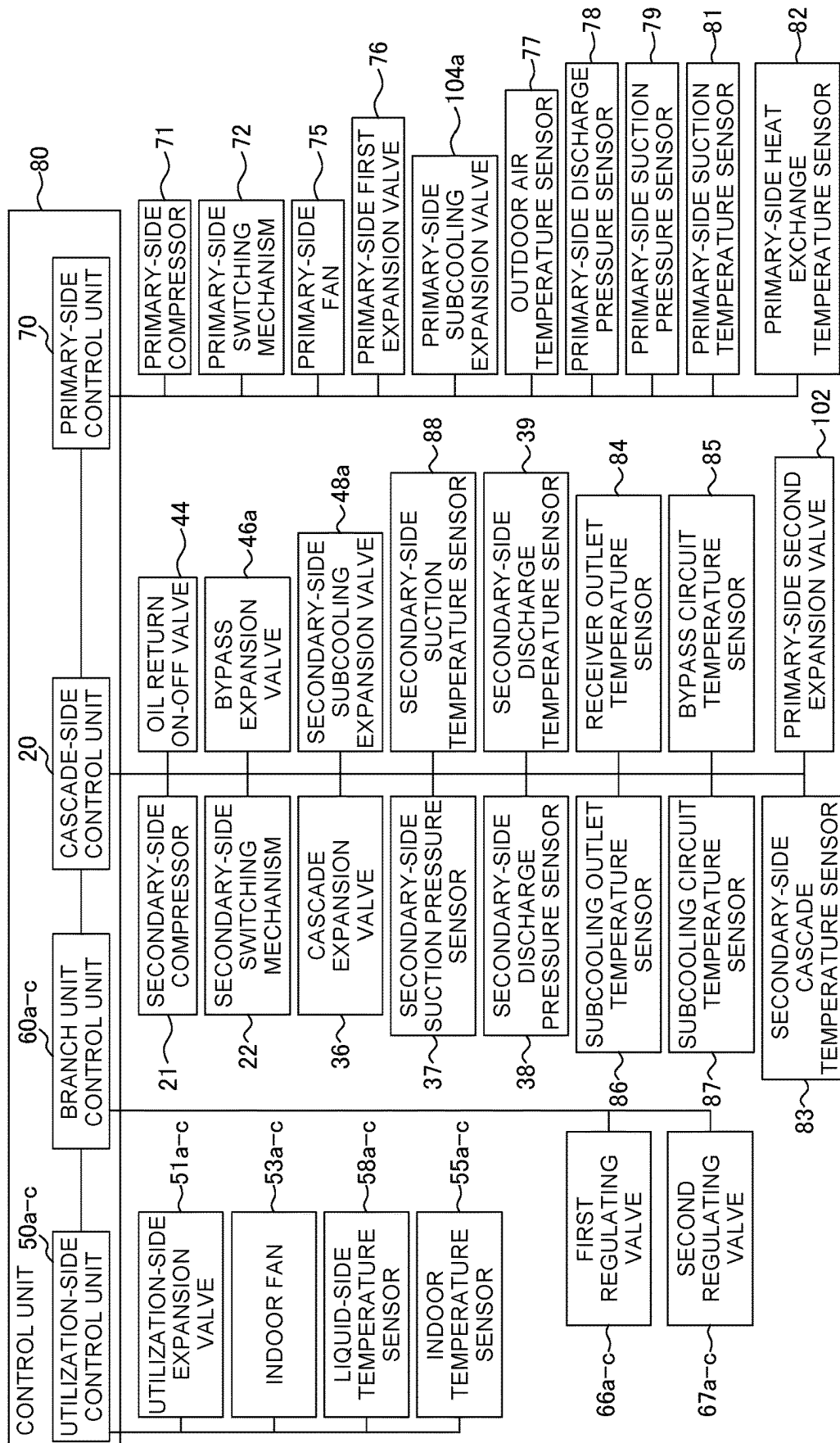


FIG. 2

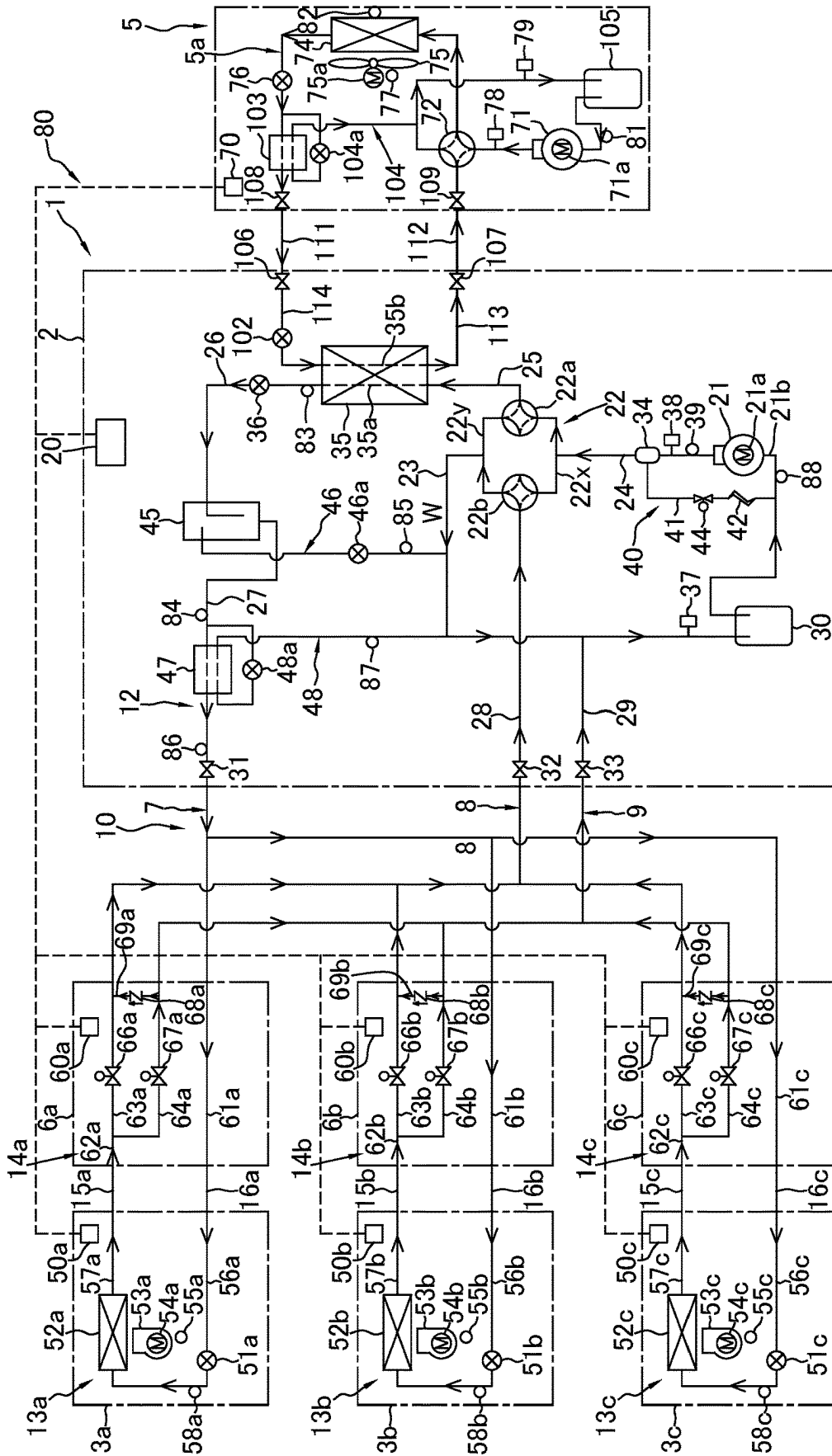


FIG. 3

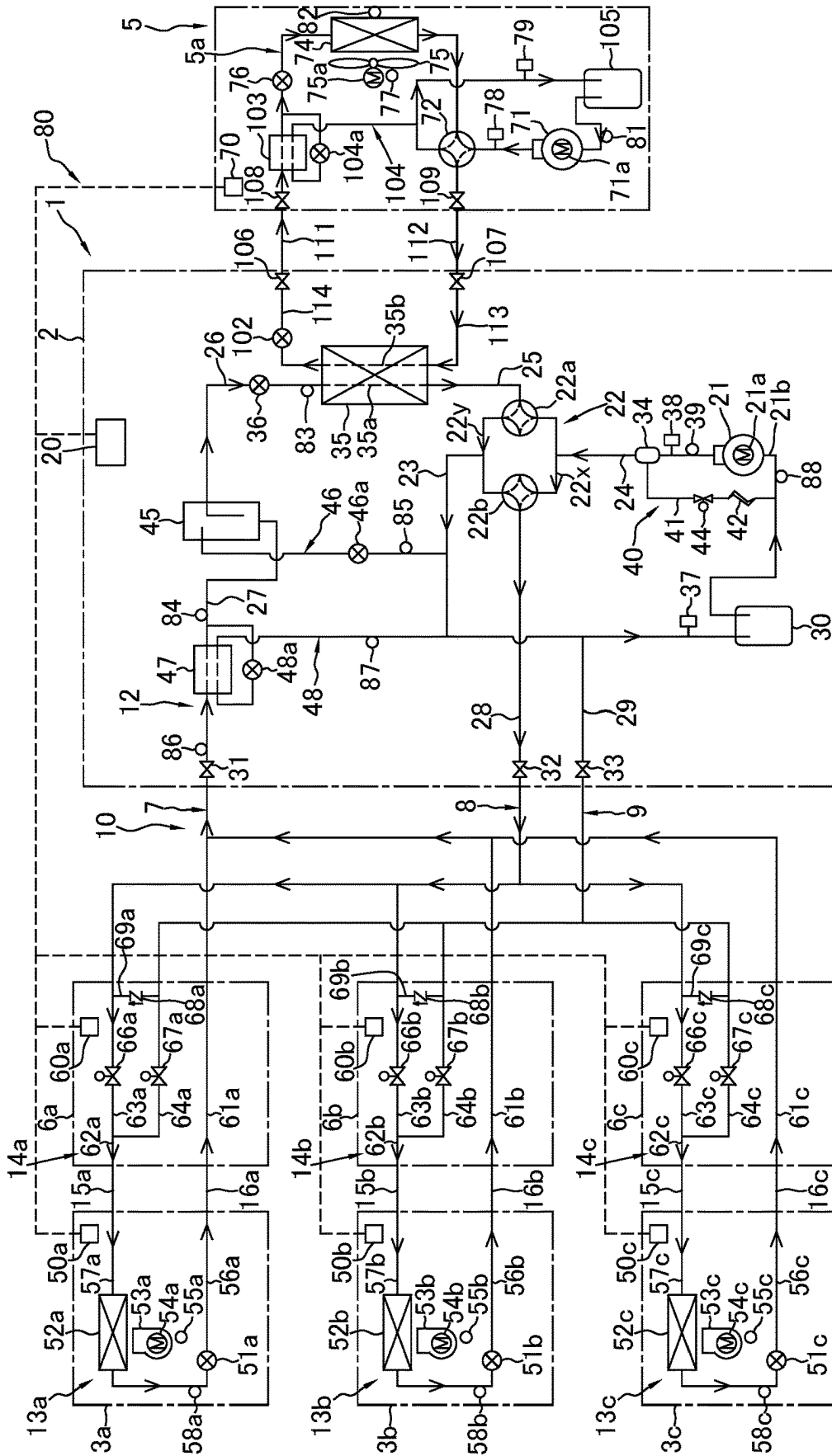


FIG. 4

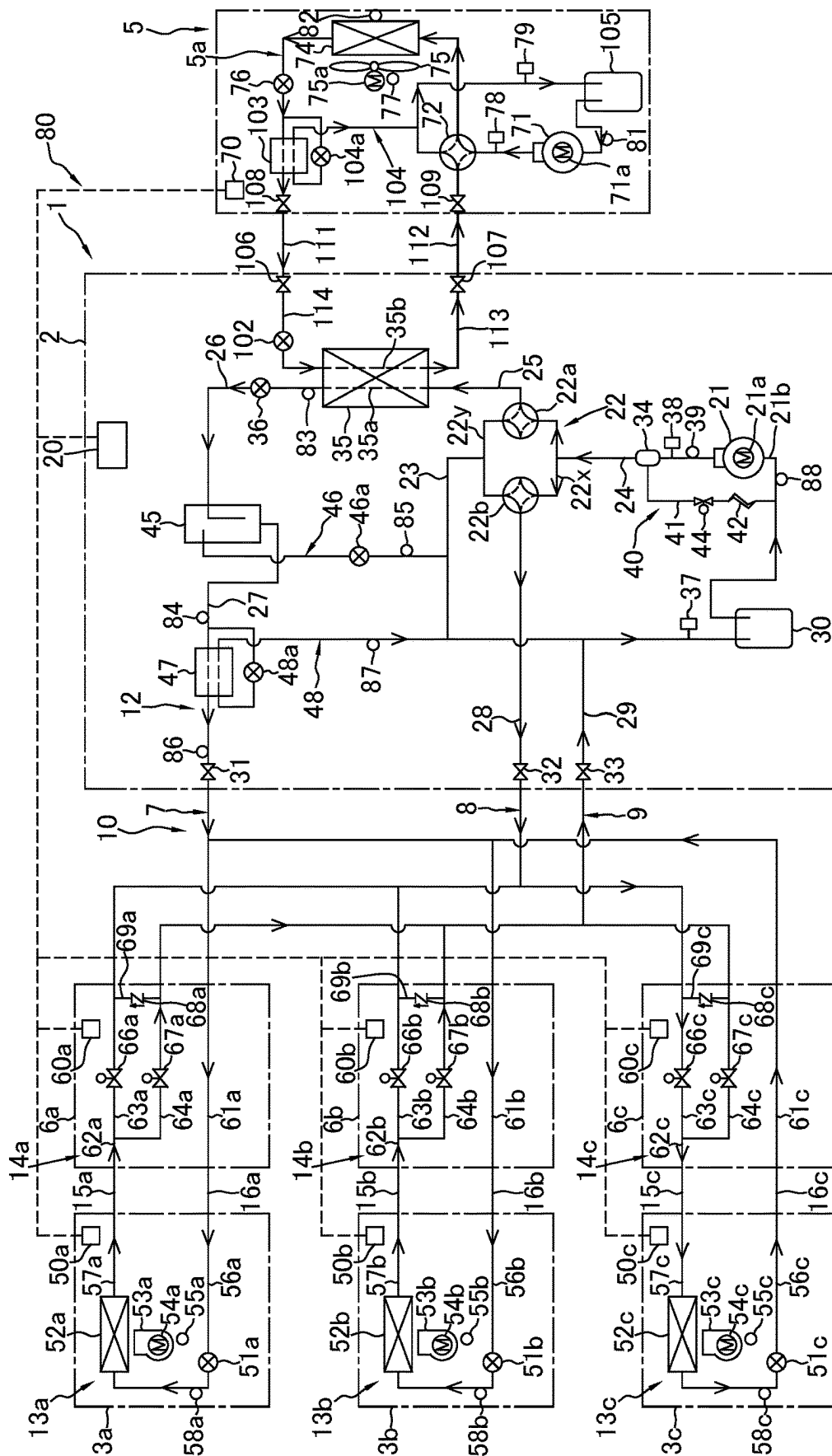


FIG. 5

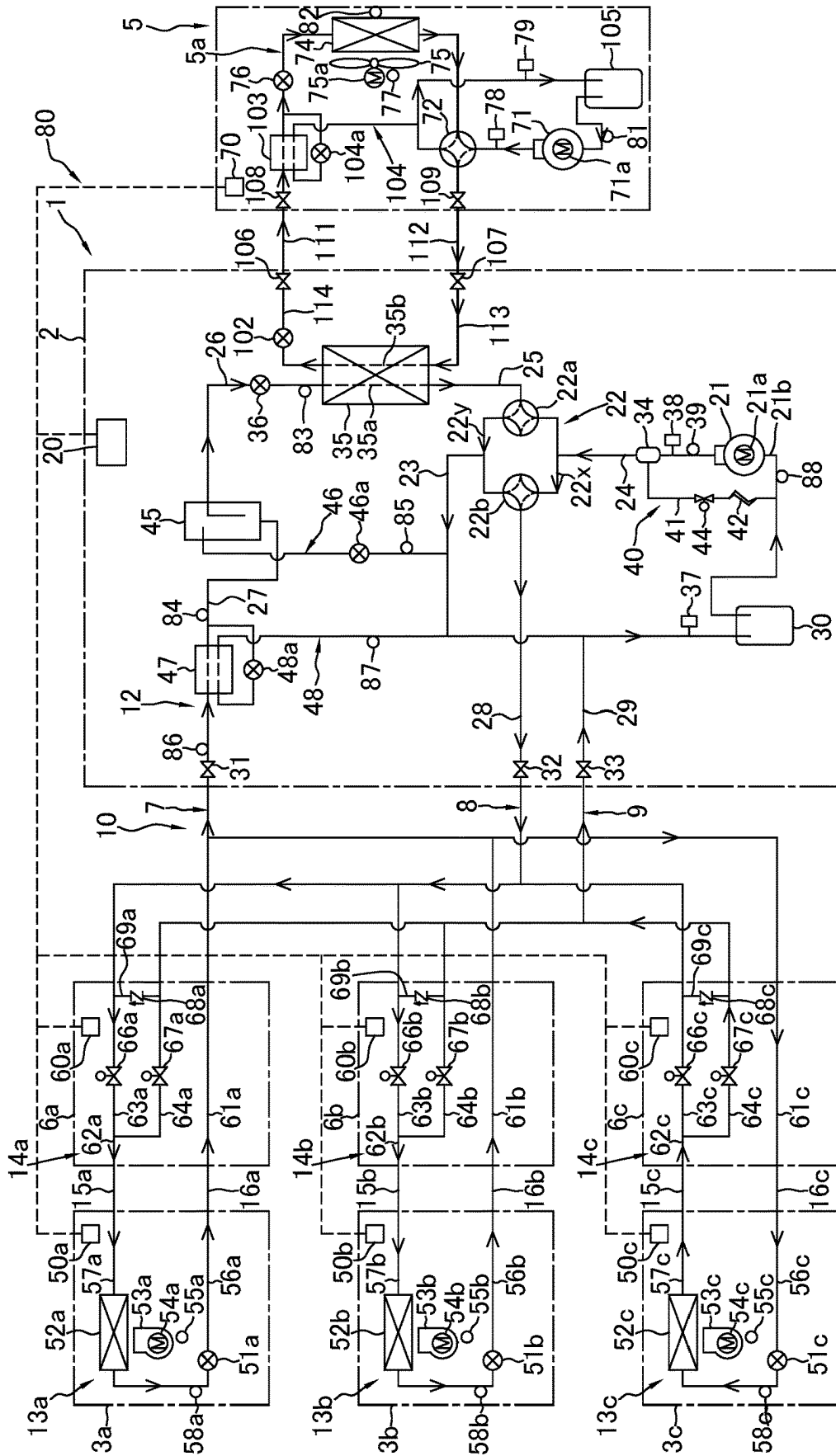


FIG. 6

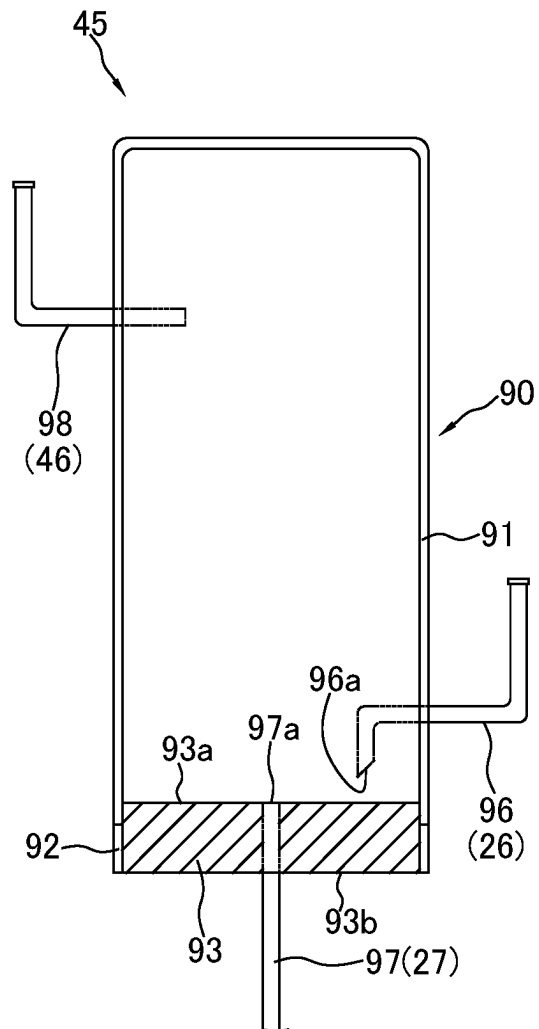


FIG. 7

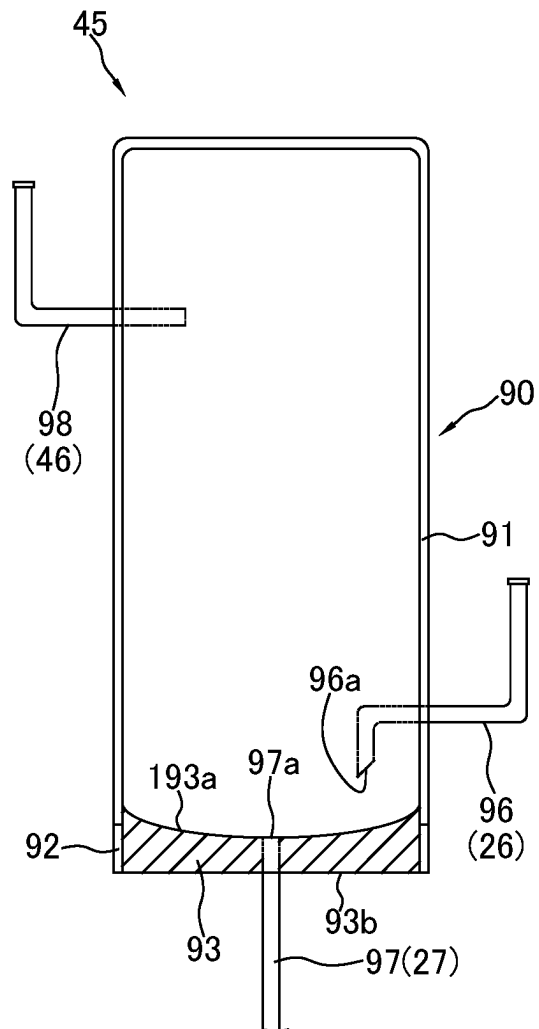


FIG. 8

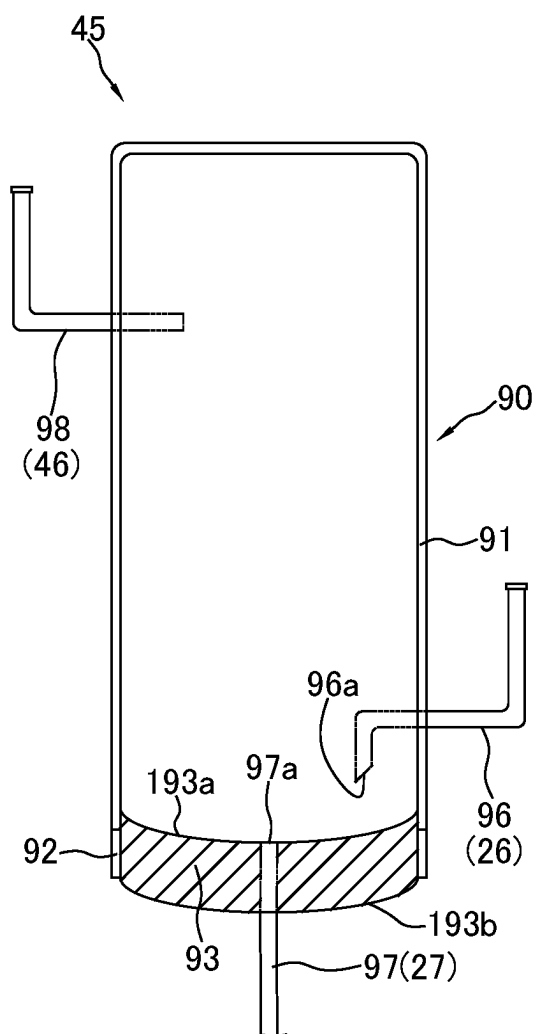


FIG. 9

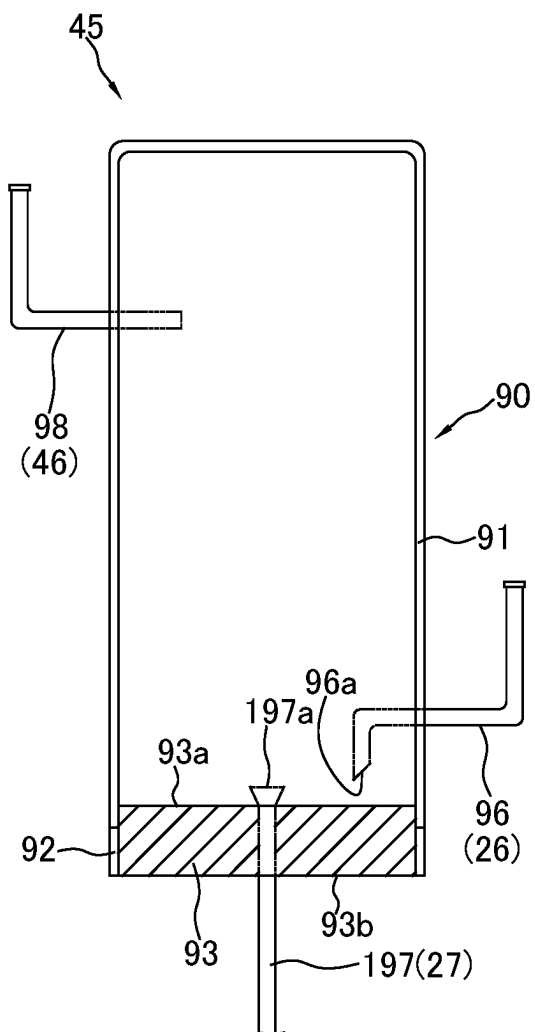


FIG. 10

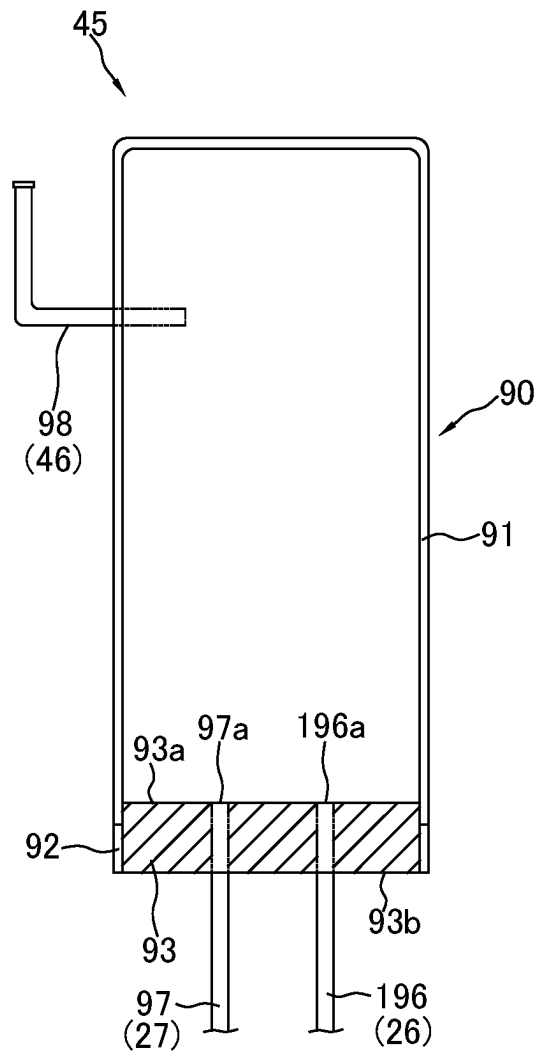


FIG. 11

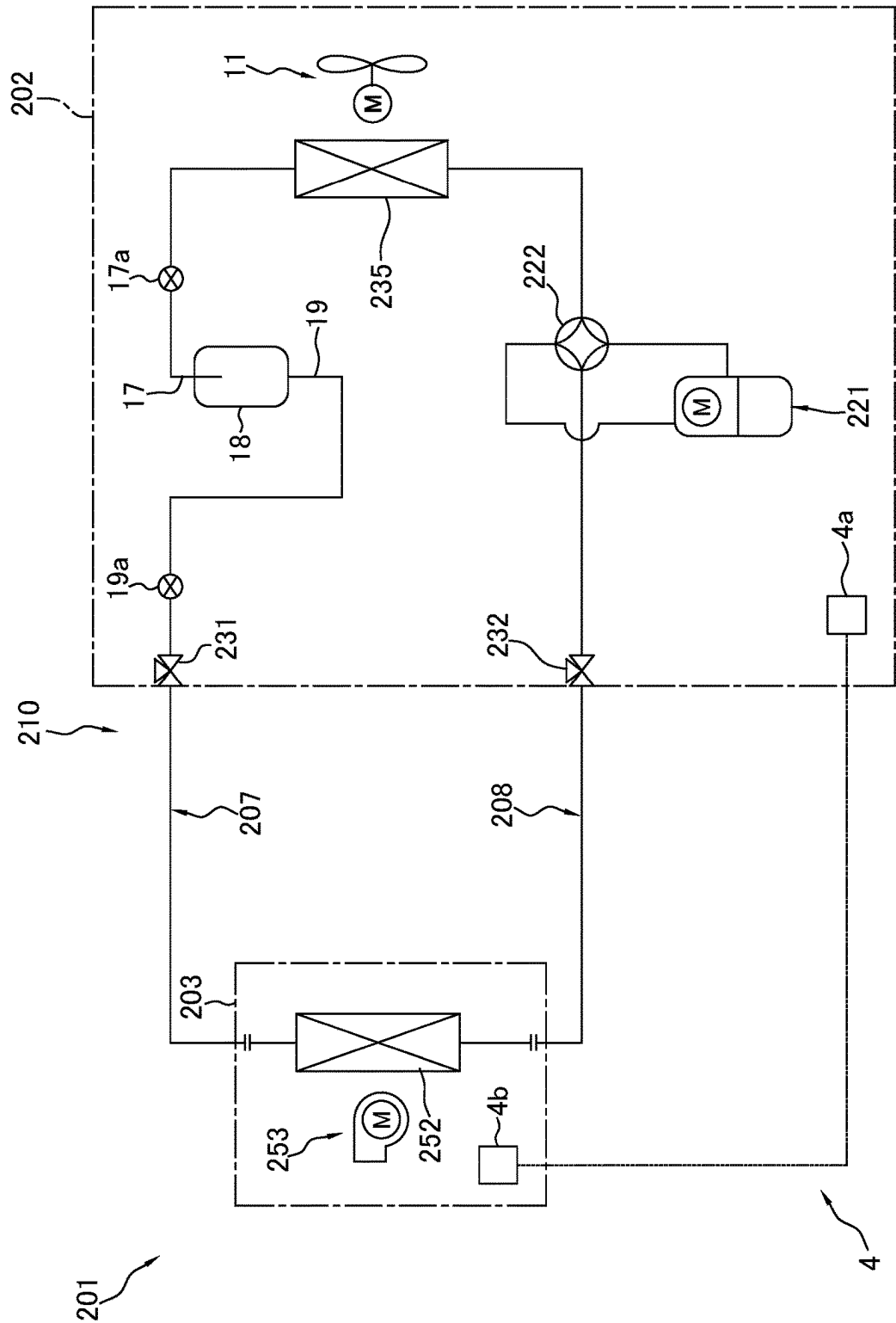


FIG. 12

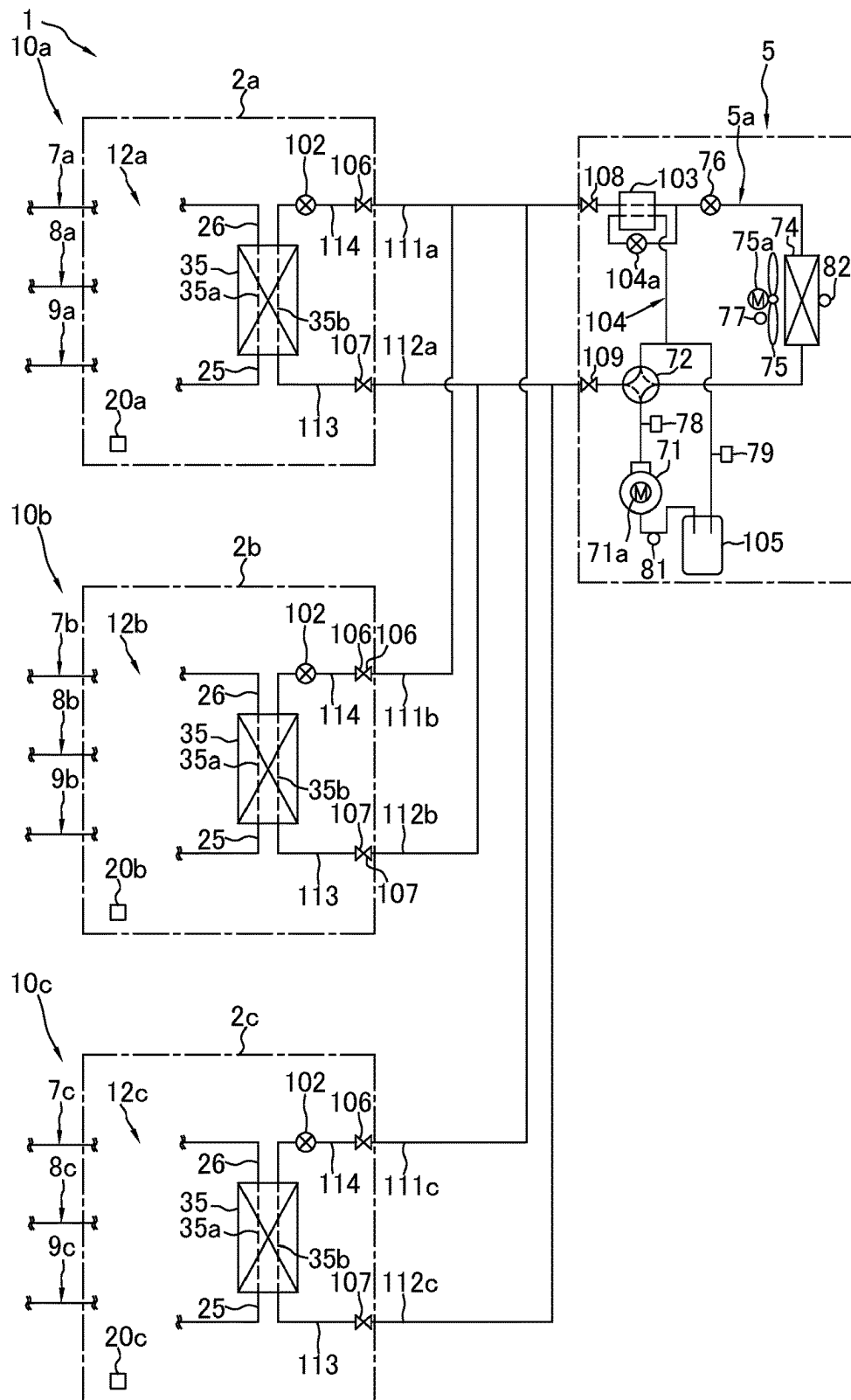


FIG. 13

1

REFRIGERANT VESSEL AND REFRIGERATION CYCLE APPARATUS

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of PCT International Application No. PCT/JP2022/035473, filed on Sep. 22, 2022, which claims priority under 35 U.S.C. § 119(a) to Patent Application No. JP 2021-161999, filed in Japan on Sep. 30, 2021, all of which are hereby expressly incorporated by reference into the present application.

The present disclosure relates to a refrigerant vessel and a refrigeration cycle apparatus.

TECHNICAL FIELD

Background Art

Conventionally, a receiver for reserving a refrigerant has been used in a refrigerant circuit included in a refrigeration cycle apparatus.

As such a refrigeration cycle apparatus, for example, a refrigeration cycle apparatus described in Patent Literature 1 (JP 2008-164225 A) uses a carbon dioxide refrigerant as a refrigerant and uses incompatible PAG oil as a refrigerating machine oil. With this refrigeration cycle apparatus, in order to avoid accumulation of refrigerating machine oil at a bottom of a refrigerant vessel such as a receiver, it is proposed that an outlet pipe connected to a side peripheral surface of the refrigerant vessel is disposed so that a vicinity of a distal end located inside the refrigerant vessel is obliquely cut and the distal end is in contact with a bottom part of the refrigerant vessel.

SUMMARY

A refrigerant vessel according to a first aspect is a refrigerant vessel that reserves a refrigerant in a refrigerant circuit. In the refrigerant circuit, the refrigerant and the refrigerating machine oil circulate. The refrigerant vessel includes a vessel body, a first refrigerant pipe, and a second refrigerant pipe. The first refrigerant pipe is connected to the vessel body. The second refrigerant pipe is connected to the vessel body. The refrigerating machine oil is incompatible with the refrigerant. The refrigerating machine oil has a higher density than the refrigerant. The second refrigerant pipe extends downward from a bottom part of the vessel body.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic configuration diagram of a refrigeration cycle apparatus.

FIG. 2 is a schematic functional block configuration diagram of the refrigeration cycle apparatus.

FIG. 3 is a diagram illustrating motion (a flow of a refrigerant) in cooling operation of the refrigeration cycle apparatus.

FIG. 4 is a diagram illustrating motion (a flow of a refrigerant) in heating operation of the refrigeration cycle apparatus.

FIG. 5 is a diagram illustrating motion (a flow of a refrigerant) in simultaneous cooling and heating operation (cooling main operation) of the refrigeration cycle apparatus.

2

FIG. 6 is a diagram illustrating motion (a flow of a refrigerant) in simultaneous cooling and heating operation (heating main operation) of the refrigeration cycle apparatus.

FIG. 7 is a schematic configuration diagram of a secondary-side receiver.

FIG. 8 is a schematic configuration diagram of a secondary-side receiver according to another embodiment A.

FIG. 9 is a schematic configuration diagram of a secondary-side receiver according to another embodiment B.

FIG. 10 is a schematic configuration diagram of a secondary-side receiver according to another embodiment C.

FIG. 11 is a schematic configuration diagram of a secondary-side receiver according to another embodiment D.

FIG. 12 is a schematic configuration diagram of a refrigerant circuit including an intermediate-pressure receiver according to another embodiment E.

FIG. 13 is a schematic configuration diagram of a refrigeration cycle apparatus according to another embodiment F.

DESCRIPTION OF EMBODIMENTS

(1) Configuration of Refrigeration Cycle Apparatus

FIG. 1 is a schematic configuration diagram of a refrigeration cycle apparatus 1. FIG. 2 is a schematic functional block configuration diagram of the refrigeration cycle apparatus 1.

The refrigeration cycle apparatus 1 is an apparatus used for cooling and heating of an indoor space in a building or the like by performing vapor compression refrigeration cycle operation.

The refrigeration cycle apparatus 1 includes a binary refrigerant circuit including a vapor compression primary-side refrigerant circuit 5a (corresponding to a first refrigerant circuit) and a vapor compression secondary-side refrigerant circuit 10 (corresponding to a refrigerant circuit, a second refrigerant circuit), and performs a binary refrigeration cycle. In the present embodiment, for example, R32 or R410A is sealed as a refrigerant in the primary-side refrigerant circuit 5a. In the secondary-side refrigerant circuit 10, for example, carbon dioxide is sealed as a refrigerant. In the secondary-side refrigerant circuit 10, a refrigerating machine oil incompatible with the refrigerant is used as a refrigerating machine oil that circulates in the circuit together with the refrigerant to improve lubricity of a sliding portion and the like. The refrigerating machine oil has a higher density and a higher specific gravity than the refrigerant in a use state, and thus tends to be located below the refrigerant. In the present embodiment, a polyalkylene glycol oil (PAG oil) that is incompatible with carbon dioxide as a refrigerant and has a higher density than the carbon dioxide refrigerant is used.

The primary-side refrigerant circuit 5a and the secondary-side refrigerant circuit 10 are thermally connected via a cascade heat exchanger 35 described later.

The refrigeration cycle apparatus 1 is configured by connecting a primary-side unit 5, a cascade unit 2, a plurality of branch units 6a, 6b, and 6c, and a plurality of utilization units 3a, 3b, and 3c to each other via pipes. The primary-side unit 5 and the cascade unit 2 are connected via a primary-side first connection pipe 111 and a primary-side second connection pipe 112. The cascade unit 2 and the plurality of branch units 6a, 6b, and 6c are connected via three connection pipes, namely, a secondary-side second connection pipe 9, a secondary-side first connection pipe 8, and a secondary-side third connection pipe 7. The plurality of branch units

3

6a, 6b, and 6c and the plurality of utilization units 3a, 3b, and 3c are connected via first connecting tubes 15a, 15b, and 15c and second connecting tubes 16a, 16b, and 16c. The present embodiment provides the single primary-side unit 5. The present embodiment provides the single cascade unit 2. The plurality of utilization units 3a, 3b, and 3c according to the present embodiment includes three utilization units, namely, the first utilization unit 3a, the second utilization unit 3b, and the third utilization unit 3c. In the present embodiment, the plurality of branch units 6a, 6b, and 6c is three branch units of the first branch unit 6a, the second branch unit 6b, and the third branch unit 6c.

In the refrigeration cycle apparatus 1, the utilization units 3a, 3b, and 3c can individually perform cooling operation or heating operation, and heat can be recovered between the utilization units by sending a refrigerant from the utilization unit performing the heating operation to the utilization unit performing the cooling operation. Specifically, heat is recovered in the present embodiment by performing cooling main operation or heating main operation of simultaneously performing cooling operation and heating operation. In addition, the refrigeration cycle apparatus 1 is configured to balance heat loads of the cascade unit 2 in accordance with entire heat loads of the plurality of utilization units 3a, 3b, and 3c also in consideration of the heat recovery (the cooling main operation or the heating main operation).

(2) Primary-Side Refrigerant Circuit

The primary-side refrigerant circuit 5a includes a primary-side compressor 71, a primary-side switching mechanism 72, a primary-side heat exchanger 74, a primary-side first expansion valve 76, a primary-side subcooling heat exchanger 103, a primary-side subcooling circuit 104, a primary-side subcooling expansion valve 104a, a first liquid shutoff valve 108, the primary-side first connection pipe 111, a second liquid shutoff valve 106, the second refrigerant pipe 114, a primary-side second expansion valve 102, the cascade heat exchanger 35 shared with the secondary-side refrigerant circuit 10, a first refrigerant pipe 113, a second gas shutoff valve 107, the primary-side second connection pipe 112, a first gas shutoff valve 109, and a primary-side accumulator 105. This primary-side refrigerant circuit 5a specifically includes a primary-side flow path 35b of the cascade heat exchanger 35.

The primary-side compressor 71 is a device for compressing a primary-side refrigerant, and includes, for example, a scroll type or other positive-displacement compressor whose operating capacity can be varied by controlling an inverter for a compressor motor 71a.

The primary-side accumulator 105 is provided at a half-way portion of the suction flow path connecting the primary-side switching mechanism 72 and a suction side of the primary-side compressor 71.

When the cascade heat exchanger 35 functions as an evaporator for the primary-side refrigerant, the primary-side switching mechanism 72 enters a fifth connection state of connecting the suction side of the primary-side compressor 71 and a gas side of a primary-side flow path 35b of the cascade heat exchanger 35 (see solid lines in the primary-side switching mechanism 72 in FIG. 1). In addition, when the cascade heat exchanger 35 functions as a radiator for the primary-side refrigerant, the primary-side switching mechanism 72 enters a sixth connection state of connecting a discharge side of the primary-side compressor 71 and the gas side of the primary-side flow path 35b of the cascade heat exchanger 35 (see broken lines in the primary-side switching

4

mechanism 72 in FIG. 1). In such a manner, the primary-side switching mechanism 72 is a device that can switch the flow path of the refrigerant in the primary-side refrigerant circuit 5a, and includes, for example, a four-way switching valve. Then, by changing a switching state of the primary-side switching mechanism 72, the cascade heat exchanger 35 can function as the evaporator or the radiator for the primary-side refrigerant.

The cascade heat exchanger 35 is a device for causing heat exchange between a refrigerant such as R32 which is a primary-side refrigerant and a refrigerant such as carbon dioxide which is a secondary-side refrigerant without mixing the refrigerants with each other. The cascade heat exchanger 35 is, for example, a plate-type heat exchanger. The cascade heat exchanger 35 includes a secondary-side flow path 35a belonging to the secondary-side refrigerant circuit 10 and the primary-side flow path 35b belonging to the primary-side refrigerant circuit 5a. The secondary-side flow path 35a has a gas side connected to a secondary-side switching mechanism 22 via a third pipe 25, and a liquid side connected to a cascade expansion valve 36 via a fourth pipe 26. The primary-side flow path 35b has a gas side connected to the primary-side compressor 71 via the first refrigerant pipe 113, the second gas shutoff valve 107, the primary-side second connection pipe 112, the first gas shutoff valve 109, and the primary-side switching mechanism 72, and has a liquid side connected to the second refrigerant pipe 114 provided with the primary-side second expansion valve 102.

The primary-side heat exchanger 74 is a device for exchanging heat between the primary-side refrigerant and outdoor air. The primary-side heat exchanger 74 has a gas side connected to a pipe extending from the primary-side switching mechanism 72. The primary-side heat exchanger 74 includes, for example, a fin-and-tube heat exchanger including a large number of heat transfer tubes and fins.

The primary-side first expansion valve 76 is provided on a liquid pipe extending from a liquid side of the primary-side heat exchanger 74 to the primary-side subcooling heat exchanger 103. The primary-side first expansion valve 76 is an electrically powered expansion valve that has an adjustable opening degree for adjusting a flow rate of the primary-side refrigerant flowing in a portion on a liquid side of the primary-side refrigerant circuit 5a.

The primary-side subcooling circuit 104 branches from a portion between the primary-side first expansion valve 76 and the primary-side subcooling heat exchanger 103, and is connected to a portion between the primary-side switching mechanism 72 and the primary-side accumulator 105 on the suction flow path. The primary-side subcooling expansion valve 104a is an electrically powered expansion valve that is provided upstream of the primary-side subcooling heat exchanger 103 in the primary-side subcooling circuit 104 and has an adjustable opening degree for adjusting the flow rate of the primary-side refrigerant.

The primary-side subcooling heat exchanger 103 causes heat exchange between a refrigerant flowing from the primary-side first expansion valve 76 toward the first liquid shutoff valve 108 and a refrigerant decompressed at the primary-side subcooling expansion valve 104a in the primary-side subcooling circuit 104.

The primary-side first connection pipe 111 is a pipe connecting the first liquid shutoff valve 108 and the second liquid shutoff valve 106, and connects the primary-side unit 5 and the cascade unit 2.

5

The primary-side second connection pipe 112 is a pipe connecting the first gas shutoff valve 109 and the second gas shutoff valve 107, and connects the primary-side unit 5 and the cascade unit 2.

The second refrigerant pipe 114 is a pipe extending from a liquid side of the primary-side flow path 35b of the cascade heat exchanger 35 to the second liquid shutoff valve 106.

The primary-side second expansion valve 102 is provided on the second refrigerant pipe 114. The primary-side second expansion valve 102 is an electric expansion valve that has an adjustable opening degree for adjusting a flow rate of the primary-side refrigerant flowing in the primary-side flow path 35b of the cascade heat exchanger 35.

The first refrigerant pipe 113 is a pipe extending from the gas side of the primary-side flow path 35b of the cascade heat exchanger 35 to the second gas shutoff valve 107.

The first gas shutoff valve 109 is provided between the primary-side second connection pipe 112 and the primary-side switching mechanism 72.

(3) Secondary-Side Refrigerant Circuit

The secondary-side refrigerant circuit 10 includes the plurality of utilization units 3a, 3b, and 3c, the plurality of branch units 6a, 6b, and 6c, and the cascade unit 2, which are connected to each other. Each of the utilization units 3a, 3b, and 3c is connected to a corresponding one of the branch units 6a, 6b, and 6c one on one. Specifically, the utilization unit 3a and the branch unit 6a are connected via the first connecting tube 15a and the second connecting tube 16a, the utilization unit 3b and the branch unit 6b are connected via the first connecting tube 15b and the second connecting tube 16b, and the utilization unit 3c and the branch unit 6c are connected via the first connecting tube 15c and the second connecting tube 16c. Each of the branch units 6a, 6b, and 6c are connected to the cascade unit 2 via three connection pipes, namely, the secondary-side third connection pipe 7, the secondary-side first connection pipe 8, and the secondary-side second connection pipe 9. Specifically, the secondary-side third connection pipe 7, the secondary-side first connection pipe 8, and the secondary-side second connection pipe 9 extending from the cascade unit 2 are each branched into a plurality of pipes connected to the branch units 6a, 6b, and 6c.

In accordance with an operating state, either the refrigerant in a gas-liquid two-phase state or the refrigerant in a gas state flows in the secondary-side first connection pipe 8. Note that, in accordance with the operating state, the refrigerant in a supercritical state flows in the secondary-side first connection pipe 8. In accordance with the operating state, either the refrigerant in the gas-liquid two-phase state or the refrigerant in the gas state flows in the secondary-side second connection pipe 9. In accordance with the operating state, either the refrigerant in the gas-liquid two-phase state or the refrigerant in a liquid state flows in the secondary-side third connection pipe 7. Note that, in accordance with the operating state, the refrigerant in a supercritical state flows in the secondary-side third connection pipe 7.

The secondary-side refrigerant circuit 10 includes a cascade circuit 12, branch circuits 14a, 14b, and 14c, and utilization circuits 13a, 13b, and 13c, which are connected to each other.

The cascade circuit 12 principally includes a secondary-side compressor 21, the secondary-side switching mechanism 22, a first pipe 28, a second pipe 29, a suction flow path 23, a discharge flow path 24, the third pipe 25, the fourth pipe 26, a fifth pipe 27, the cascade heat exchanger 35, the

6

cascade expansion valve 36, a third shutoff valve 31, a first shutoff valve 32, a second shutoff valve 33, a secondary-side accumulator 30, an oil separator 34, an oil return circuit 40, a secondary-side receiver 45 (corresponding to a refrigerant vessel), a bypass circuit 46, a bypass expansion valve 46a, a secondary-side subcooling heat exchanger 47, a secondary-side subcooling circuit 48, and a secondary-side subcooling expansion valve 48a. The cascade circuit 12 of the secondary-side refrigerant circuit 10 specifically includes the secondary-side flow path 35a of the cascade heat exchanger 35.

The secondary-side compressor 21 is a device for compressing the secondary-side refrigerant, and is exemplarily constituted by a scroll type or other positive-displacement compressor whose operating capacity can be varied by controlling an inverter for a compressor motor 21a. The secondary-side compressor 21 is controlled in accordance with an operating load so as to have larger operating capacity as the load increases.

The secondary-side switching mechanism 22 is a mechanism that can switch a connection state of the secondary-side refrigerant circuit 10, specifically, the flow path of the refrigerant in the cascade circuit 12. In the present embodiment, the secondary-side switching mechanism 22 includes a discharge-side connection portion 22x, a suction-side connection portion 22y, a first switching valve 22a, and a second switching valve 22b. An end of the discharge flow path 24 on a side opposite to the secondary-side compressor 21 is connected to the discharge-side connection portion 22x. An end of the suction flow path 23 on a side opposite to the secondary-side compressor 21 is connected to the suction-side connection portion 22y. The first switching valve 22a and the second switching valve 22b are provided in parallel to each other between the discharge flow path 24 and the suction flow path 23 of the secondary-side compressor 21. The first switching valve 22a is connected to one end of the discharge-side connection portion 22x and one end of the suction-side connection portion 22y. The second switching valve 22b is connected to the other end of the discharge-side connection portion 22x and the other end of the suction-side connection portion 22y. In the present embodiment, each of the first switching valve 22a and the second switching valve 22b includes a four-way switching valve. Each of the first switching valve 22a and the second switching valve 22b has four connection ports, namely, a first connection port, a second connection port, a third connection port, and a fourth connection port. In the first switching valve 22a and the second switching valve 22b according to the present embodiment, each of the fourth ports is a closed connection port not connected to the flow path of the secondary-side refrigerant circuit 10. In the first switching valve 22a, the first connection port is connected to the one end of the discharge-side connection portion 22x, the second connection port is connected to the third pipe 25 extending from the secondary-side flow path 35a of the cascade heat exchanger 35, and the third connection port is connected to the one end of the suction-side connection portion 22y. The first switching valve 22a switches between a switching state in which the first connection port and the second connection port are connected and the third connection port and the fourth connection port are connected and a switching state in which the third connection port and the second connection port are connected and the first connection port and the fourth connection port are connected. The second switching valve 22b has the first connection port connected to the other end of the discharge-side connection portion 22x, the second connection port connected to the first

pipe 28, and the third connection port connected to the other end of the suction-side connection portion 22y. The second switching valve 22b switches between a switching state in which the first connection port and the second connection port are connected and the third connection port and the fourth connection port are connected and a switching state in which the third connection port and the second connection port are connected and the first connection port and the fourth connection port are connected.

When the secondary-side refrigerant discharged from the secondary-side compressor 21 is prevented from being sent to the secondary-side first connection pipe 8 while the cascade heat exchanger 35 functions as a radiator for the secondary-side refrigerant, the secondary-side switching mechanism 22 is switched to a first connection state in which the discharge flow path 24 and the third pipe 25 are connected by the first switching valve 22a and the first pipe 28 and the suction flow path 23 are connected by the second switching valve 22b. The first connection state of the secondary-side switching mechanism 22 is a connection state adopted during the cooling operation described later. When the cascade heat exchanger 35 functions as an evaporator for the secondary-side refrigerant, the secondary-side switching mechanism 22 is switched to a second connection state in which the discharge flow path 24 and the first pipe 28 are connected by the second switching valve 22b and the third pipe 25 and the suction flow path 23 are connected by the first switching valve 22a. The second connection state of the secondary-side switching mechanism 22 is a connection state adopted during the heating operation and during the heating main operation described later. When the secondary-side refrigerant discharged from the secondary-side compressor 21 is sent to the secondary-side first connection pipe 8 while the cascade heat exchanger 35 functions as a radiator for the secondary-side refrigerant, the secondary-side switching mechanism 22 is switched to a third connection state in which the discharge flow path 24 and the third pipe 25 are connected by the first switching valve 22a and the discharge flow path 24 and the first pipe 28 are connected by the second switching valve 22b. The third connection state of the secondary-side switching mechanism 22 is a connection state adopted during the cooling main operation described later.

As described above, the cascade heat exchanger 35 is a device for causing heat exchange between the refrigerant such as R32 which is the primary-side refrigerant and the refrigerant such as carbon dioxide which is the secondary-side refrigerant without mixing the refrigerants with each other. The cascade heat exchanger 35 includes the secondary-side flow path 35a in which the secondary-side refrigerant in the secondary-side refrigerant circuit 10 flows and the primary-side flow path 35b in which the primary-side refrigerant in the primary-side refrigerant circuit 5a flows, so as to be shared between the primary-side unit 5 and the cascade unit 2. Note that, in the present embodiment, the cascade heat exchanger 35 is disposed inside a cascade casing (not illustrated) of the cascade unit 2. The gas side of the primary-side flow path 35b of the cascade heat exchanger 35 extends to the primary-side second connection pipe 112 outside the cascade casing via the first refrigerant pipe 113 and the second gas shutoff valve 107. The liquid side of the primary-side flow path 35b of the cascade heat exchanger 35 extends to the primary-side first connection pipe 111 outside the cascade casing via the second refrigerant pipe 114 provided with the primary-side second expansion valve 102 and the second liquid shutoff valve 106.

The cascade expansion valve 36 is an expansion valve for adjusting a flow rate of the secondary-side refrigerant flowing in the cascade heat exchanger 35. The cascade expansion valve 36 is an electric expansion valve that is connected to a liquid side of the cascade heat exchanger 35 and has an adjustable opening degree. The cascade expansion valve 36 is provided on the fourth pipe 26.

Each of the third shutoff valve 31, the first shutoff valve 32, and the second shutoff valve 33 is provided at a connecting port with an external device or pipe (specifically, the connection pipe 7, 8, or 9). Specifically, the third shutoff valve 31 is connected to the secondary-side third connection pipe 7 led out of the cascade unit 2. The first shutoff valve 32 is connected to the secondary-side first connection pipe 8 led out of the cascade unit 2. The second shutoff valve 33 is connected to the secondary-side second connection pipe 9 led out of the cascade unit 2.

The first pipe 28 is a refrigerant pipe connecting the first shutoff valve 32 and the secondary-side switching mechanism 22. Specifically, the first pipe 28 connects the first shutoff valve 32 and the second connection port of the second switching valve 22b of the secondary-side switching mechanism 22.

The suction flow path 23 connects the secondary-side switching mechanism 22 and a suction side of the secondary-side compressor 21. Specifically, the suction flow path 23 connects the suction-side connection portion 22y of the secondary-side switching mechanism 22 and the suction side of the secondary-side compressor 21. The secondary-side accumulator 30 is provided at a halfway portion of the suction flow path 23.

The second pipe 29 is a refrigerant pipe that connects the second shutoff valve 33 to a halfway portion of the suction flow path 23. In the present embodiment, the second pipe 29 is connected to the suction flow path 23 at a connection point of the suction flow path 23 between the suction-side connection portion 22y of the secondary-side switching mechanism 22 and the secondary-side accumulator 30.

The discharge flow path 24 is a refrigerant pipe connecting a discharge side of the secondary-side compressor 21 and the secondary-side switching mechanism 22. Specifically, the discharge flow path 24 connects the discharge side of the secondary-side compressor 21 and the discharge-side connection portion 22x of the secondary-side switching mechanism 22.

The third pipe 25 is a refrigerant pipe connecting the secondary-side switching mechanism 22 and a gas side of the cascade heat exchanger 35. Specifically, the third pipe 25 connects the second connection port of the first switching valve 22a of the secondary-side switching mechanism 22 and a gas-side end of the secondary-side flow path 35a in the cascade heat exchanger 35.

The fourth pipe 26 is a refrigerant pipe connecting the liquid side (opposite to the gas side, and opposite to the side provided with the secondary-side switching mechanism 22) of the cascade heat exchanger 35 and the secondary-side receiver 45. Specifically, the fourth pipe 26 connects a liquid side end (opposite to the gas side) of the secondary-side flow path 35a in the cascade heat exchanger 35 and the secondary-side receiver 45.

The secondary-side receiver 45 is a refrigerant vessel that reserves a residue refrigerant in the secondary-side refrigerant circuit 10. Extended from the secondary-side receiver 45 are the fourth pipe 26, the fifth pipe 27, and the bypass circuit 46.

The bypass circuit 46 is a refrigerant pipe connecting a gas phase region which is an upper region in the secondary-

side receiver 45 and the suction flow path 23. Specifically, the bypass circuit 46 is connected between the secondary-side switching mechanism 22 and the secondary-side accumulator 30 on the suction flow path 23. The bypass circuit 46 is provided with the bypass expansion valve 46a. The bypass expansion valve 46a is an electrically powered expansion valve that can adjust a quantity of the refrigerant guided from inside the secondary-side receiver 45 to the suction side of the secondary-side compressor 21 by adjusting an opening degree.

The fifth pipe 27 is a refrigerant pipe connecting the secondary-side receiver 45 and the third shutoff valve 31.

The secondary-side subcooling circuit 48 is a refrigerant pipe connecting a part of the fifth pipe 27 and the suction flow path 23. Specifically, the secondary-side subcooling circuit 48 is connected between the secondary-side switching mechanism 22 and the secondary-side accumulator 30 on the suction flow path 23. In the present embodiment, the secondary-side subcooling circuit 48 extends to branch from a portion between the secondary-side receiver 45 and the secondary-side subcooling heat exchanger 47.

The secondary-side subcooling heat exchanger 47 is a heat exchanger that causes heat exchange between the refrigerant flowing in a flow path belonging to the fifth pipe 27 and the refrigerant flowing in a flow path belonging to the secondary-side subcooling circuit 48. In the present embodiment, the secondary-side subcooling heat exchanger 47 is provided between a portion from where the secondary-side subcooling circuit 48 branches and the third shutoff valve 31 on the fifth pipe 27. The secondary-side subcooling expansion valve 48a is provided between a portion branching from the fifth pipe 27 and the secondary-side subcooling heat exchanger 47 on the secondary-side subcooling circuit 48. The secondary-side subcooling expansion valve 48a is an electrically powered expansion valve that has an adjustable opening degree and supplies the secondary-side subcooling heat exchanger 47 with a decompressed refrigerant.

The secondary-side accumulator 30 is a vessel that can reserve the secondary-side refrigerant, and is provided on the suction side of the secondary-side compressor 21.

The oil separator 34 is provided at a halfway portion of the discharge flow path 24. The oil separator 34 is a device for separating refrigerating machine oil discharged from the secondary-side compressor 21 along with the secondary-side refrigerant from the secondary-side refrigerant and return the refrigerating machine oil to the secondary-side compressor 21.

The oil return circuit 40 is provided to connect the oil separator 34 and the suction flow path 23. The oil return circuit 40 includes an oil return flow path 41 which is a flow path extending from the oil separator 34 and extending to join a portion between the secondary-side accumulator 30 and the suction side of the secondary-side compressor 21 on the suction flow path 23. An oil return capillary tube 42 and an oil return on-off valve 44 are provided at a halfway portion of the oil return flow path 41. When the oil return on-off valve 44 is controlled into an opened state, the refrigerating machine oil separated in the oil separator 34 passes through the oil return capillary tube 42 on the oil return flow path 41 and is returned to the suction side of the secondary-side compressor 21. In the present embodiment, when the secondary-side compressor 21 is in an operating state on the secondary-side refrigerant circuit 10, the oil return on-off valve 44 is kept in the opened state for predetermined time and is kept in a closed state for predetermined time repetitively, to control a returned quantity of the refrigerating machine oil through the oil return circuit

40. In the present embodiment, the oil return on-off valve 44 is an electromagnetic valve controlled to be opened and closed. Alternatively, the oil return on-off valve 44 may be an electrically powered expansion valve having an adjustable opening degree and not provided with the oil return capillary tube 42.

Hereinafter, the utilization circuits 13a, 13b, and 13c will be described. Since the utilization circuits 13b and 13c are configured similarly to the utilization circuit 13a, elements of the utilization circuits 13b and 13c will not be described repeatedly, assuming that a subscript "b" or "c" will replace a subscript "a" in reference signs denoting elements of the utilization circuit 13a.

The utilization circuit 13a principally includes a utilization-side heat exchanger 52a, a first utilization pipe 57a, a second utilization pipe 56a, and a utilization-side expansion valve 51a.

The utilization-side heat exchanger 52a is a device for causing heat exchange between a refrigerant and indoor air, and includes, for example, a fin-and-tube heat exchanger including a large number of heat transfer tubes and fins. The plurality of utilization-side heat exchangers 52a, 52b, and 52c are connected in parallel to the secondary-side switching mechanism 22, the suction flow path 23, and the cascade heat exchanger 35.

The second utilization pipe 56a has one end connected to a liquid side (opposite to a gas side) of the utilization-side heat exchanger 52a in the first utilization unit 3a. The other end of the second utilization pipe 56a is connected to the second connecting tube 16a. The utilization-side expansion valve 51a described above is provided at a halfway portion of the second utilization pipe 56a.

The utilization-side expansion valve 51a is an electrically powered expansion valve that has an adjustable opening degree for adjusting a flow rate of the refrigerant flowing in the utilization-side heat exchanger 52a. The utilization-side expansion valve 51a is provided on the second utilization pipe 56a.

The first utilization pipe 57a has one end connected to the gas side of the utilization-side heat exchanger 52a in the first utilization unit 3a. In the present embodiment, the first utilization pipe 57a is connected to a portion opposite to the utilization-side expansion valve 51a of the utilization-side heat exchanger 52a. The first utilization pipe 57a has the other end connected to the first connecting tube 15a.

Hereinafter, the branch circuits 14a, 14b, and 14c will be described. Since the branch circuits 14b and 14c are configured similarly to the branch circuit 14a, elements of the branch circuits 14b and 14c will not be described repeatedly, assuming that a subscript "b" or "c" will replace a subscript "a" in reference signs denoting elements of the branch circuit 14a.

The branch circuit 14a mainly includes a junction pipe 62a, a first branch pipe 63a, a second branch pipe 64a, a first regulating valve 66a, a second regulating valve 67a, a bypass pipe 69a, a check valve 68a, and a third branch pipe 61a.

The junction pipe 62a has one end connected to the first connecting tube 15a. The junction pipe 62a has the other end branched to be connected with the first branch pipe 63a and the second branch pipe 64a.

The first branch pipe 63a has a portion opposite to the junction pipe 62 and connected to the secondary-side first connection pipe 8. The first branch pipe 63a is provided with the openable and closable first regulating valve 66a.

The second branch pipe 64a has a portion opposite to the junction pipe 62 and connected to the secondary-side second

11

connection pipe 9. The second branch pipe 64a is provided with the openable and closable second regulating valve 67a.

The bypass pipe 69a is a refrigerant pipe that connects a portion of the first branch pipe 63a closer to the secondary-side first connection pipe 8 than the first regulating valve 66a and a portion of the second branch pipe 64a closer to the secondary-side second connection pipe 9 than the second regulating valve 67a. The check valve 68a is provided at a halfway portion of the bypass pipe 69a. The check valve 68a allows only a refrigerant flow from the second branch pipe 64a toward the first branch pipe 63a, and does not allow a refrigerant flow from the first branch pipe 63a toward the second branch pipe 64a.

The third branch pipe 61a has one end connected to the second connecting tube 16a. The other end of the third branch pipe 61a is connected to the secondary-side third connection pipe 7.

The first branch unit 6a can function as follows by closing the first regulating valve 66a and opening the second regulating valve 67a when the cooling operation described later is performed. The first branch unit 6a sends a refrigerant flowing into the third branch pipe 61a through the secondary-side third connection pipe 7 to the second connecting tube 16a. The refrigerant flowing in the second utilization pipe 56a in the first utilization unit 3a through the second connecting tube 16a is sent to the utilization-side heat exchanger 52a in the first utilization unit 3a through the utilization-side expansion valve 51a. The refrigerant sent to the utilization-side heat exchanger 52a is evaporated by heat exchange with indoor air, and then flows in the first connecting tube 15a via the first utilization pipe 57a. The refrigerant having flowed through the first connecting tube 15a is sent to the junction pipe 62a of the first branch unit 6a. The refrigerant having flowed through the junction pipe 62a does not flow toward the first branch pipe 63a but flows toward the second branch pipe 64a. The refrigerant flowing in the second branch pipe 64a passes through the second regulating valve 67a. A part of the refrigerant that has passed through the second regulating valve 67a is sent to the secondary-side second connection pipe 9. The remaining part of the refrigerant that has passed through the second regulating valve 67a flows so as to branch into the bypass pipe 69a provided with the check valve 68a, passes through a part of the first branch pipe 63a, and then is sent to the secondary-side first connection pipe 8. As a result, it is possible to increase a total flow path sectional area when the secondary-side refrigerant in a gas state evaporated in the utilization-side heat exchanger 52a is sent to the secondary-side compressor 21, so that a pressure loss can be reduced.

When the first utilization unit 3a cools an indoor space at the time of performing the cooling main operation and the heating main operation described later, the first branch unit 6a can function as follows by closing the first regulating valve 66a and opening the second regulating valve 67a. The first branch unit 6a sends a refrigerant flowing into the third branch pipe 61a through the secondary-side third connection pipe 7 to the second connecting tube 16a. The refrigerant flowing in the second utilization pipe 56a in the first utilization unit 3a through the second connecting tube 16a is sent to the utilization-side heat exchanger 52a in the first utilization unit 3a through the utilization-side expansion valve 51a. The refrigerant sent to the utilization-side heat exchanger 52a is evaporated by heat exchange with indoor air, and then flows in the first connecting tube 15a via the first utilization pipe 57a. The refrigerant having flowed through the first connecting tube 15a is sent to the junction pipe 62a of the first branch unit 6a. The refrigerant having

12

flowed through the junction pipe 62a flows to the second branch pipe 64a, passes through the second regulating valve 67a, and then is sent to the secondary-side second connection pipe 9.

The first branch unit 6a can function as follows by closing the second regulating valve 67a and opening the first regulating valve 66a when the heating operation described later is performed. In the first branch unit 6a, the refrigerant flowing into the first branch pipe 63a through the secondary-side first connection pipe 8 passes through the first regulating valve 66a and is sent to the junction pipe 62a. The refrigerant having flowed through the junction pipe 62a flows in the first utilization pipe 57a in the utilization unit 3a via the first connecting tube 15a and is sent to the utilization-side heat exchanger 52a. The refrigerant sent to the utilization-side heat exchanger 52a radiates heat through heat exchange with indoor air, and then passes through the utilization-side expansion valve 51a provided on the second utilization pipe 56a. The refrigerant having passed through the second utilization pipe 56a flows through the third branch pipe 61a of the first branch unit 6a via the second connecting tube 16a, and then is sent to the secondary-side third connection pipe 7.

When the first utilization unit 3a heats an indoor space at the time of performing the cooling main operation and the heating main operation described later, the first branch unit 6a can function as follows by closing the second regulating valve 67a and opening the first regulating valve 66a. In the first branch unit 6a, the refrigerant flowing into the first branch pipe 63a through the secondary-side first connection pipe 8 passes through the first regulating valve 66a and is sent to the junction pipe 62a. The refrigerant having flowed through the junction pipe 62a flows in the first utilization pipe 57a in the utilization unit 3a via the first connecting tube 15a and is sent to the utilization-side heat exchanger 52a. The refrigerant sent to the utilization-side heat exchanger 52a radiates heat through heat exchange with indoor air, and then passes through the utilization-side expansion valve 51a provided on the second utilization pipe 56a. The refrigerant having passed through the second utilization pipe 56a flows through the third branch pipe 61a of the first branch unit 6a via the second connecting tube 16a, and then is sent to the secondary-side third connection pipe 7.

The first branch unit 6a, as well as the second branch unit 6b and the third branch unit 6c, similarly have such a function. Accordingly, the first branch unit 6a, the second branch unit 6b, and the third branch unit 6c can individually switchably cause the utilization-side heat exchangers 52a, 52b, and 52c to function as a refrigerant evaporator or a refrigerant radiator.

(4) Primary-Side Unit

The primary-side unit 5 is disposed in a space different from a space provided with the utilization units 3a, 3b, and 3c and the branch units 6a, 6b, and 6c, on a roof, or the like.

The primary-side unit 5 includes a part of the primary-side refrigerant circuit 5a described above, a primary-side fan 75, various sensors, and a primary-side control unit 70, and a primary-side casing (not illustrated).

The primary-side unit 5 includes, as a part of the primary-side refrigerant circuit 5a, the primary-side compressor 71, the primary-side switching mechanism 72, the primary-side heat exchanger 74, the primary-side first expansion valve 76, the primary-side subcooling heat exchanger 103, the primary-side subcooling circuit 104, the primary-side subcool-

13

ing expansion valve **104a**, the first liquid shutoff valve **108**, the first gas shutoff valve **109**, and the primary-side accumulator **105** in the primary-side casing.

The primary-side fan **75** is provided in the primary-side unit **5**, and generates an air flow of guiding outdoor air into the primary-side heat exchanger **74**, and exhausting, to outdoors, air obtained after heat exchange with the primary-side refrigerant flowing in the primary-side heat exchanger **74**. The primary-side fan **75** is driven by a primary-side fan motor **75a**.

The primary-side unit **5** is provided with the various sensors. Specifically, there are provided an outdoor air temperature sensor **77** that detects a temperature of outdoor air before passing through the primary-side heat exchanger **74**, a primary-side discharge pressure sensor **78** that detects a pressure of the primary-side refrigerant discharged from the primary-side compressor **71**, a primary-side suction pressure sensor **79** that detects a pressure of the primary-side refrigerant sucked into the primary-side compressor **71**, a primary-side suction temperature sensor **81** that detects a temperature of the primary-side refrigerant sucked into the primary-side compressor **71**, and a primary-side heat exchange temperature sensor **82** that detects a temperature of the refrigerant flowing in the primary-side heat exchanger **74**.

The primary-side control unit **70** controls motion of the elements **71** (**71a**), **72**, **75** (**75a**), **76**, and **104a** provided in the primary-side unit **5**. The primary-side control unit **70** includes a processor such as a CPU or a microcomputer provided to control the primary-side unit **5** and a memory, so as to transmit and receive control signals and the like to and from a remote controller (not illustrated), and to transmit and receive control signals and the like between a cascade-side control unit **20** in a cascade unit **2**, branch unit control units **60a**, **60b**, and **60c**, and utilization-side control units **50a**, **50b**, and **50c**.

(5) Cascade Unit

The cascade unit **2** is disposed in a space different from a space provided with the utilization units **3a**, **3b**, and **3c** and the branch units **6a**, **6b**, and **6c**, on a roof, or the like.

The cascade unit **2** is connected to the branch units **6a**, **6b**, and **6c** via the connection pipes **7**, **8**, and **9**, to constitute a part of the secondary-side refrigerant circuit **10**. The cascade unit **2** is connected to the primary-side unit **5** via the primary-side first connection pipe **111** and the primary-side second connection pipe **112**, to constitute a part of the primary-side refrigerant circuit **5a**.

The cascade unit **2** mainly includes the cascade circuit **12** described above, various sensors, the cascade-side control unit **20**, the second liquid shutoff valve **106**, the second refrigerant pipe **114**, the primary-side second expansion valve **102**, the first refrigerant pipe **113**, and the second gas shutoff valve **107** that constitute a part of the primary-side refrigerant circuit **5a**, the cascade casing (not illustrated), and the like.

The cascade unit **2** is provided with a secondary-side suction pressure sensor **37** that detects a pressure of the secondary-side refrigerant on the suction side of the secondary-side compressor **21**, a secondary-side discharge pressure sensor **38** that detects a pressure of the secondary-side refrigerant on the discharge side of the secondary-side compressor **21**, a secondary-side discharge temperature sensor **39** that detects a temperature of the secondary-side refrigerant on the discharge side of the secondary-side compressor **21**, a secondary-side suction temperature sensor

14

88 that detects a temperature of the secondary-side refrigerant on the suction side of the secondary-side compressor **21**, a secondary-side cascade temperature sensor **83** that detects a temperature of the secondary-side refrigerant flowing between the secondary-side flow path **35a** of the cascade heat exchanger **35** and the cascade expansion valve **36**, a receiver outlet temperature sensor **84** that detects a temperature of the secondary-side refrigerant flowing between the secondary-side receiver **45** and the secondary-side subcooling heat exchanger **47**, a bypass circuit temperature sensor **85** that detects a temperature of the secondary-side refrigerant flowing downstream of the bypass expansion valve **46a** in the bypass circuit **46**, a subcooling outlet temperature sensor **86** that detects a temperature of the secondary-side refrigerant flowing between the secondary-side subcooling heat exchanger **47** and the third shutoff valve **31**, and a subcooling circuit temperature sensor **87** that detects a temperature of the secondary-side refrigerant flowing at an outlet of the secondary-side subcooling heat exchanger **47** in the secondary-side subcooling circuit **48**.

The cascade-side control unit **20** controls motion of the elements **21** (**21a**), **22**, **36**, **44**, **46a**, **48a**, and **102** provided in the cascade casing of the cascade unit **2**. The cascade-side control unit **20** includes a processor such as a CPU or a microcomputer provided to control the cascade unit **2** and a memory, so as to transmit and receive control signals and the like between the primary-side control unit **70** in the primary-side unit **5**, the utilization-side control units **50a**, **50b**, and **50c** in the utilization units **3a**, **3b**, and **3c**, and the branch unit control units **60a**, **60b**, and **60c**.

In such a manner, the cascade-side control unit **20** can control not only the elements constituting the cascade circuit **12** of the secondary-side refrigerant circuit **10** but also the primary-side second expansion valve **102** constituting a part of the primary-side refrigerant circuit **5a**. Therefore, the cascade-side control unit **20** controls a valve opening degree of the primary-side second expansion valve **102** on the basis of a condition of the cascade circuit **12** controlled by the cascade-side control unit **20**, so as to bring the condition of the cascade circuit **12** closer to a desired condition. Specifically, it is possible to control an amount of heat received by the secondary-side refrigerant flowing in the secondary-side flow path **35a** of the cascade heat exchanger **35** in the cascade circuit **12** from the primary-side refrigerant flowing in the primary-side flow path **35b** of the cascade heat exchanger **35** or an amount of heat given by the secondary-side refrigerant to the primary-side refrigerant.

(6) Utilization Unit

The utilization units **3a**, **3b**, and **3c** are installed by being embedded in or being suspended from a ceiling on an indoor space of a building or the like, or by being hung on a wall surface in the indoor space, or the like.

The utilization units **3a**, **3b**, and **3c** are connected to the cascade unit **2** via the connection pipes **7**, **8**, and **9**.

The utilization units **3a**, **3b**, and **3c** respectively include the utilization circuits **13a**, **13b**, and **13c** constituting a part of the secondary-side refrigerant circuit **10**.

Hereinafter, the utilization units **3a**, **3b**, and **3c** will be described in terms of their configurations. The second utilization unit **3b** and the third utilization unit **3c** are configured similarly to the first utilization unit **3a**. The configuration of only the first utilization unit **3a** will thus be described here. As for the configuration of each of the second utilization unit **3b** and the third utilization unit **3c**, elements will be denoted by reference signs obtained by

15

replacing a subscript “a” in reference signs of elements of the first utilization unit **3a** with a subscript “b” or “c”, and these elements will not be described repeatedly.

The first utilization unit **3a** mainly includes the utilization circuit **13a** described above, an indoor fan **53a**, the utilization-side control unit **50a**, and various sensors. Note that the indoor fan **53a** includes an indoor fan motor **54a**.

The indoor fan **53a** generates an air flow by sucking indoor air into the unit and supplying the indoor space with supply air obtained after heat exchange with the refrigerant flowing in the utilization-side heat exchanger **52a**. The indoor fan **53a** is driven by the indoor fan motor **54a**.

The utilization unit **3a** is provided with a liquid-side temperature sensor **58a** that detects a temperature of a refrigerant on the liquid side of the utilization-side heat exchanger **52a**. The utilization unit **3a** is further provided with an indoor temperature sensor **55a** that detects an indoor temperature as temperature of air introduced from the indoor space before passing through the utilization-side heat exchanger **52a**.

The utilization-side control unit **50a** controls motion of the elements **51a** and **53a** (**54a**) constituting the utilization unit **3a**. The utilization-side control unit **50a** includes a processor such as a CPU or a microcomputer provided to control the utilization unit **3a** and a memory, so as to transmit and receive control signals and the like to and from the remote controller (not illustrated), and to transmit and receive control signals and the like between the cascade-side control unit **20** in the cascade unit **2**, the branch unit control units **60a**, **60b**, and **60c**, and the primary-side control unit **70** in the primary-side unit **5**.

Note that the second utilization unit **3b** includes the utilization circuit **13b**, an indoor fan **53b**, the utilization-side control unit **50b**, and an indoor fan motor **54b**. The third utilization unit **3c** includes the utilization circuit **13c**, an indoor fan **53c**, the utilization-side control unit **50c**, and an indoor fan motor **54c**.

(7) Branch Unit

The branch units **6a**, **6b**, and **6c** are installed in a space above a ceiling of an indoor space of a building or the like.

Each of the branch units **6a**, **6b**, and **6c** is connected to a corresponding one of the utilization units **3a**, **3b**, and **3c** one by one. The branch units **6a**, **6b**, and **6c** are connected to the cascade unit **2** via the connection pipes **7**, **8**, and **9**.

Next, the branch units **6a**, **6b**, and **6c** will be described next in terms of their configurations. The second branch unit **6b** and the third branch unit **6c** are configured similarly to the first branch unit **6a**. The configuration of only the first branch unit **6a** will thus be described here. As for the configuration of each of the second branch unit **6b** and the third branch unit **6c**, elements will be denoted by reference signs obtained by replacing a subscript “a” in reference signs of elements of the first branch unit **6a** with a subscript “b” or “c”, and these elements will not be described repeatedly.

The first branch unit **6a** mainly includes the branch circuit **14a** described above and the branch unit control unit **60a**.

The branch unit control unit **60a** controls motion of the elements **66a** and **67a** constituting the branch unit **6a**. The branch unit control unit **60a** includes a processor such as a CPU or a microcomputer provided to control the branch unit **6a** and a memory, so as to transmit and receive control signals and the like to and from the remote controller (not depicted), and to transmit and receive control signals and the like between the cascade-side control unit **20** in the cascade

16

unit **2**, the utilization units **3a**, **3b**, and **3c**, and the primary-side control unit **70** in the primary-side unit **5**.

Note that the second branch unit **6b** includes the branch circuit **14b** and the branch unit control unit **60b**. The third branch unit **6c** includes the branch circuit **14c** and the branch unit control unit **60c**.

(8) Control Unit

In the refrigeration cycle apparatus **1**, the heat cascade-side control unit **20**, the utilization-side control units **50a**, **50b**, and **50c**, the branch unit control units **60a**, **60b**, and **60c**, and the primary-side control unit **70** described above are communicably connected to each other in a wired or wireless manner to constitute a control unit **80**. Therefore, the control unit **80** controls motion of the elements **21**(**21a**), **22**, **36**, **44**, **46a**, **48a**, **51a**, **51b**, **51c**, **53a**, **53b**, **53c** (**54a**, **54b**, **54c**), **66a**, **66b**, **66c**, **67a**, **67b**, **67c**, **71** (**71a**), **72**, **75** (**75a**), **76**, **104a** on the basis of detection information of various sensors **37**, **38**, **39**, **83**, **84**, **85**, **86**, **87**, **88**, **77**, **78**, **79**, **81**, **82**, **58a**, **58b**, **58c**, and the like, and instruction information received from a remote controller (not illustrated) and the like.

(9) Motion of Refrigeration Cycle Apparatus

Next, motion of the refrigeration cycle apparatus **1** will be described with reference to FIGS. **3** to **6**.

The refrigeration cycle operation of the refrigeration cycle apparatus **1** can be mainly divided into cooling operation, heating operation, cooling main operation, and heating main operation.

Here, the cooling operation is refrigeration cycle operation in which only the utilization unit in which the utilization-side heat exchanger functions as a refrigerant evaporator exists, and the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant with respect to an evaporation load of the entire utilization unit.

Here, the heating operation is refrigeration cycle operation in which only the utilization unit in which the utilization-side heat exchanger functions as a refrigerant radiator exists, and the cascade heat exchanger **35** functions as an evaporator for the secondary-side refrigerant with respect to a radiation load of the entire utilization unit.

The cooling main operation is operation in which the utilization unit in which the utilization-side heat exchanger functions as a refrigerant evaporator and the utilization unit in which the utilization-side heat exchanger functions as a refrigerant radiator are mixed. The cooling main operation is refrigeration cycle operation in which, when an evaporation load is a main heat load of the entire utilization unit, the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant in order to process the evaporation load of the entire utilization unit.

The heating main operation is operation in which the utilization unit in which the utilization-side heat exchanger functions as a refrigerant evaporator and the utilization unit in which the utilization-side heat exchanger functions as a refrigerant radiator are mixed. The heating main operation is refrigeration cycle operation in which, when a radiation load is a main heat load of the entire utilization unit, the cascade heat exchanger **35** functions as an evaporator for the secondary-side refrigerant in order to process the radiation load of the entire utilization unit.

Note that the motion of the refrigeration cycle apparatus **1** including the refrigeration cycle operation is performed by the control unit **80** described above.

During the cooling operation, for example, all of the utilization-side heat exchangers **52a**, **52b**, and **52c** in the utilization units **3a**, **3b**, and **3c** functions as a refrigerant evaporator, and the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant. In the cooling operation, the primary-side refrigerant circuit **5a** and the secondary-side refrigerant circuit **10** of the refrigeration cycle apparatus **1** are configured as illustrated in FIG. 3. Note that arrows attached to the primary-side refrigerant circuit **5a** and arrows attached to the secondary-side refrigerant circuit **10** in FIG. 3 indicate flows of the refrigerant during the cooling operation.

Specifically, in the primary-side unit **5**, the primary-side switching mechanism **72** is switched to the fifth connection state to cause the cascade heat exchanger **35** to function as an evaporator for the primary-side refrigerant. The fifth connection state of the primary-side switching mechanism **72** is depicted by solid lines in the primary-side switching mechanism **72** in FIG. 3. Accordingly, in the primary-side unit **5**, the primary-side refrigerant discharged from the primary-side compressor **71** passes through the primary-side switching mechanism **72** and exchanges heat with outdoor air supplied from the primary-side fan **75** in the primary-side heat exchanger **74** to be condensed. The primary-side refrigerant condensed in the primary-side heat exchanger **74** passes through the primary-side first expansion valve **76** controlled into a fully opened state, and a part of the refrigerant flows toward the first liquid shutoff valve **108** through the primary-side subcooling heat exchanger **103**, and another part of the refrigerant branches into the primary-side subcooling circuit **104**. The refrigerant flowing in the primary-side subcooling circuit **104** is decompressed when passing through the primary-side subcooling expansion valve **104a**. The refrigerant flowing from the primary-side first expansion valve **76** toward the first liquid shutoff valve **108** exchanges heat with the refrigerant decompressed by the primary-side subcooling expansion valve **104a** and flowing in the primary-side subcooling circuit **104** in the primary-side subcooling heat exchanger **103**, and is cooled until reaching a subcooled state. The refrigerant in the subcooled state flows through the primary-side first connection pipe **111**, the second liquid shutoff valve **106**, and the second refrigerant pipe **114** in that order, and is decompressed when passing through the primary-side second expansion valve **102**. Here, a valve opening degree of the primary-side second expansion valve **102** is controlled such that a degree of superheating of the primary-side refrigerant sucked into the primary-side compressor **71** satisfies a predetermined condition. When flowing in the primary-side flow path **35b** of the cascade heat exchanger **35**, the primary-side refrigerant decompressed by the primary-side second expansion valve **102** evaporates by exchanging heat with the secondary-side refrigerant in through the secondary-side flow path **35a**, and flows toward the second gas shutoff valve **107** through the first refrigerant pipe **113**. The refrigerant having passed through the second gas shutoff valve **107** passes through the primary-side second connection pipe **112** and the first gas shutoff valve **109**, and then reaches the primary-side switching mechanism **72**. The refrigerant having passed through the primary-side switching mechanism **72** joins the refrigerant having flowed through the primary-side subcooling circuit **104**, and is then sucked into the primary-side compressor **71** via the primary-side accumulator **105**.

In the cascade unit **2**, by switching the secondary-side switching mechanism **22** to the first connection state, the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant. In the first connection state of the secondary-side switching mechanism **22**, the discharge flow path **24** and the third pipe **25** are connected by the first switching valve **22a**, and the first pipe **28** and the suction flow path **23** are connected by the second switching valve **22b**. In the first to third utilization units **3a**, **3b**, **3c**, the second regulating valves **67a**, **67b**, **67c** are controlled to the opened state. Accordingly, all of the utilization-side heat exchangers **52a**, **52b**, and **52c** in the utilization units **3a**, **3b**, and **3c** function as refrigerant evaporators. All of the utilization-side heat exchangers **52a**, **52b**, and **52c** of the utilization units **3a**, **3b**, and **3c** and the suction side of the secondary-side compressor **21** of the cascade unit **2** are connected via the first utilization pipes **57a**, **57b**, and **57c**, the first connecting tubes **15a**, **15b**, and **15c**, the junction pipes **62a**, **62b**, and **62c**, the second branch pipes **64a**, **64b**, and **64c**, the bypass pipes **69a**, **69b**, and **69c**, a part of the first branch pipes **63a**, **63b**, and **63c**, the secondary-side first connection pipe **8**, and the secondary-side second connection pipe **9**. In addition, an opening degree of the secondary-side subcooling expansion valve **48a** is controlled such that a degree of subcooling of the secondary-side refrigerant flowing through the outlet of the secondary-side subcooling heat exchanger **47** toward the secondary-side third connection pipe **7** satisfies a predetermined condition. The bypass expansion valve **46a** is controlled to the closed state. In the utilization units **3a**, **3b**, and **3c**, the opening degrees of the utilization-side expansion valves **51a**, **51b**, and **51c** are adjusted.

In the cooling operation, the secondary-side refrigerant circuit **10** controls capacity, for example, by controlling a frequency of the secondary-side compressor **21** so that evaporation temperature of the secondary-side refrigerant in the utilization-side heat exchangers **52a**, **52b**, and **52c** becomes a predetermined secondary-side evaporation target temperature. The opening degree of the cascade expansion valve **36** is adjusted such that the secondary-side refrigerant flowing in the cascade heat exchanger **35** has a critical pressure or less. The primary-side refrigerant circuit **5a** controls capacity, for example, by controlling a frequency of the primary-side compressor **71** such that evaporation temperature of the primary-side refrigerant in the primary-side flow path **35b** of the cascade heat exchanger **35** becomes a predetermined primary-side evaporation target temperature. In such a manner, in the cooling operation, either or both of the control for increasing the valve opening degree of the cascade expansion valve **36** and the control for increasing the frequency of the primary-side compressor **71** in the primary-side refrigerant circuit **5a** are executed, and thus, the carbon dioxide refrigerant flowing in the cascade heat exchanger **35** is controlled so as not to exceed a critical point.

In such a secondary-side refrigerant circuit **10**, a secondary-side high-pressure refrigerant compressed and discharged by the secondary-side compressor **21** is sent to the secondary-side flow path **35a** of the cascade heat exchanger **35** through the first switching valve **22a** of the secondary-side switching mechanism **22**. The secondary-side high-pressure refrigerant flowing in the secondary-side flow path **35a** of the cascade heat exchanger **35** radiates heat, and the primary-side refrigerant flowing in the primary-side flow path **35b** of the cascade heat exchanger **35** is evaporated. The secondary-side refrigerant having radiated heat in the cascade heat exchanger **35** passes through the cascade expansion

sion valve 36 whose opening degree is adjusted, and then flows into the secondary-side receiver 45. A part of the refrigerant flowing out of the secondary-side receiver 45 branches and flows into the secondary-side subcooling circuit 48, is decompressed in the secondary-side subcooling expansion valve 48a, and then joins the suction flow path 23. In the secondary-side subcooling heat exchanger 47, another part of the refrigerant having flowed out of the secondary-side receiver 45 is cooled by the refrigerant flowing in the secondary-side subcooling circuit 48, and is then sent to the secondary-side third connection pipe 7 through the third shutoff valve 31.

Then, the refrigerant sent to the secondary-side third connection pipe 7 is branched into three portions to pass through the third branch pipes 61a, 61b, and 61c of the first to third branch units 6a, 6b, and 6c. Thereafter, the refrigerant having flowed through the second connecting tubes 16a, 16b, and 16c is sent to the second utilization pipes 56a, 56b, and 56c of the first to third utilization units 3a, 3b, and 3c. The refrigerant sent to the second utilization pipes 56a, 56b, and 56c is sent to the utilization-side expansion valves 51a, 51b, and 51c in the utilization units 3a, 3b, and 3c.

Then, the refrigerant having passed through the utilization-side expansion valves 51a, 51b, and 51c whose opening degrees are adjusted exchanges heat with indoor air supplied by the indoor fans 53a, 53b, and 53c in the utilization-side heat exchangers 52a, 52b, and 52c. The refrigerant flowing in the utilization-side heat exchangers 52a, 52b, and 52c is thus evaporated into a low-pressure gas refrigerant. The indoor air is cooled and is supplied into the indoor space. The indoor space is thus cooled. The low-pressure gas refrigerant evaporated in the utilization-side heat exchangers 52a, 52b, and 52c flows in the first utilization pipes 57a, 57b, and 57c, flows through the first connecting tubes 15a, 15b, and 15c, and then is sent to the junction pipes 62a, 62b, and 62c of the first to third branch units 6a, 6b, and 6c.

Then, the low-pressure gas refrigerant sent to the junction pipes 62a, 62b, and 62c flows to the second branch pipes 64a, 64b, and 64c. A part of the refrigerant that has passed through the second regulating valves 67a, 67b, and 67c in the second branch pipes 64a, 64b, and 64c is sent to the secondary-side second connection pipe 9. The remaining part of the refrigerant that has passed through the second regulating valves 67a, 67b, and 67c passes through the bypass pipes 69a, 69b, and 69c, flows through a part of the first branch pipes 63a, 63b, and 63c, and then is sent to the secondary-side first connection pipe 8.

Then, the low-pressure gas refrigerant sent to the secondary-side first connection pipe 8 and the secondary-side second connection pipe 9 is returned to the suction side of the secondary-side compressor 21 through the first shutoff valve 32, the second shutoff valve 33, the first pipe 28, the second pipe 29, the second switching valve 22b of the secondary-side switching mechanism 22, the suction flow path 23, and the secondary-side accumulator 30.

Motion during the cooling operation is performed in such a manner.

(9-2) Heating Operation

During the heating operation, each of the utilization-side heat exchangers 52a, 52b, and 52c in the utilization units 3a, 3b, and 3c functions as a refrigerant radiator. In the heating operation, the cascade heat exchanger 35 operates to function as an evaporator for the secondary-side refrigerant. In the heating operation, the primary-side refrigerant circuit 5a and the secondary-side refrigerant circuit 10 of the refrig-

eration cycle apparatus 1 are configured as illustrated in FIG. 4. Arrows attached to the primary-side refrigerant circuit 5a and arrows attached to the secondary-side refrigerant circuit 10 in FIG. 4 indicate flows of the refrigerant during the heating operation.

Specifically, in the primary-side unit 5, the primary-side switching mechanism 72 is switched to a sixth operating state to cause the cascade heat exchanger 35 to function as a radiator for the primary-side refrigerant. The sixth operating state of the primary-side switching mechanism 72 is a connection state depicted by broken lines in the primary-side switching mechanism 72 in FIG. 4. Accordingly, in the primary-side unit 5, the primary-side refrigerant discharged from the primary-side compressor 71, having passed through the primary-side switching mechanism 72 and the first gas shutoff valve 109 passes through the primary-side second connection pipe 112 and the second gas shutoff valve 107 and is sent to the primary-side flow path 35b of the cascade heat exchanger 35. The refrigerant flowing in the primary-side flow path 35b of the cascade heat exchanger 35 is condensed by exchanging heat with the secondary-side refrigerant flowing in the secondary-side flow path 35a. When flowing in the second refrigerant pipe 114, the primary-side refrigerant condensed in the cascade heat exchanger 35 passes through the primary-side second expansion valve 102 controlled to the fully opened state. The refrigerant that has passed through the primary-side second expansion valve 102 flows through the second liquid shutoff valve 106, the primary-side first connection pipe 111, the first liquid shutoff valve 108, and the primary-side subcooling heat exchanger 103 in that order, and is decompressed by the primary-side first expansion valve 76. During heating operation, the primary-side subcooling expansion valve 104a is controlled to the closed state. Accordingly, the refrigerant does not flow to the primary-side subcooling circuit 104 and does not exchange heat in the primary-side subcooling heat exchanger 103. The valve opening degree of the primary-side first expansion valve 76 is controlled such that, for example, the degree of superheating of the refrigerant sucked into the primary-side compressor 71 satisfies a predetermined condition. The refrigerant decompressed by the primary-side first expansion valve 76 evaporates by exchanging heat with outdoor air supplied from the primary-side fan 75 in the primary-side heat exchanger 74, passes through the primary-side switching mechanism 72 and the primary-side accumulator 105, and is sucked into the primary-side compressor 71.

In the cascade unit 2, the secondary-side switching mechanism 22 is switched to the second connection state. The cascade heat exchanger 35 thus functions as an evaporator for the secondary-side refrigerant. In the second connection state of the secondary-side switching mechanism 22, the discharge flow path 24 and the first pipe 28 are connected by the second switching valve 22b, and the third pipe 25 and the suction flow path 23 are connected by the first switching valve 22a. The opening degree of the cascade expansion valve 36 is adjusted. In the first to third branch units 6a, 6b, and 6c, the first regulating valves 66a, 66b, and 66c are controlled to the opened state, and the second regulating valves 67a, 67b, and 67c are controlled to the closed state. Accordingly, all of the utilization-side heat exchangers 52a, 52b, and 52c in the utilization units 3a, 3b, and 3c function as refrigerant radiators. The utilization-side heat exchangers 52a, 52b, and 52c in the utilization units 3a, 3b, and 3c and the discharge side of the secondary-side compressor 21 in the cascade unit 2 are connected via the discharge flow path 24, the first pipe 28, the secondary-side first connection pipe

21

8, the first branch pipes 63a, 63b, and 63c, the junction pipes 62a, 62b, and 62c, the first connecting tubes 15a, 15b, and 15c, and the first utilization pipes 57a, 57b, and 57c. The secondary-side subcooling expansion valve 48a and the bypass expansion valve 46a are controlled to the closed state. In the utilization units 3a, 3b, and 3c, the opening degrees of the utilization-side expansion valves 51a, 51b, and 51c are adjusted.

During the heating operation, the secondary-side refrigerant circuit 10 controls capacity on the secondary-side compressor 21 so as to achieve a frequency at which the loads in the utilization-side heat exchangers 52a, 52b, and 52c can be processed. As a result, in the heating operation, the secondary-side refrigerant discharged from the secondary-side compressor 21 is controlled to be in a critical state exceeding the critical pressure. The primary-side refrigerant circuit 5a controls capacity, for example, by controlling the frequency of the primary-side compressor 71 such that condensation temperature of the primary-side refrigerant in the primary-side flow path 35b of the cascade heat exchanger 35 becomes a predetermined primary-side condensation target temperature.

In such a secondary-side refrigerant circuit 10, the high-pressure refrigerant compressed and discharged by the secondary-side compressor 21 is sent to the first pipe 28 through the second switching valve 22b of the secondary-side switching mechanism 22. The refrigerant sent to the first pipe 28 is sent to the secondary-side first connection pipe 8 through the first shutoff valve 32.

Then, the high-pressure refrigerant sent to the secondary-side first connection pipe 8 is branched into three portions to be sent to the first branch pipes 63a, 63b, and 63c in the utilization units 3a, 3b, and 3c which are utilization units in operation. The high-pressure refrigerant sent to the first branch pipes 63a, 63b, and 63c passes through the first regulating valves 66a, 66b, and 66c, and flows in the junction pipes 62a, 62b, and 62c. Thereafter, the refrigerant having flowed in the first connecting tubes 15a, 15b, and 15c and the first utilization pipes 57a, 57b, and 57c is sent to the utilization-side heat exchangers 52a, 52b, and 52c.

Then, the high-pressure refrigerant sent to the utilization-side heat exchangers 52a, 52b, and 52c exchanges heat with indoor air supplied by the indoor fans 53a, 53b, and 53c in the utilization-side heat exchangers 52a, 52b, and 52c. The refrigerant flowing in the utilization-side heat exchangers 52a, 52b, and 52c thus radiates heat. The indoor air is heated and supplied into the indoor space. The indoor space is thus heated. The refrigerant having radiated heat in the utilization-side heat exchangers 52a, 52b, and 52c flows in the second utilization pipes 56a, 56b, and 56c and passes through the utilization-side expansion valves 51a, 51b, and 51c whose opening degrees are adjusted. The secondary-side refrigerant that has passed through the utilization-side expansion valves 51a, 51b, and 51c has the critical pressure or less. Thereafter, the refrigerant having flowed through the second connecting tubes 16a, 16b, and 16c flows in the third branch pipes 61a, 61b, and 61c of the branch units 6a, 6b, and 6c.

The refrigerant sent to the third branch pipes 61a, 61b, and 61c is sent to the secondary-side third connection pipe 7 to join.

The refrigerant sent to the secondary-side third connection pipe 7 passes through the third shutoff valve 31 and then is sent to the cascade expansion valve 36. The flow rate of the refrigerant sent to the cascade expansion valve 36 is adjusted at the cascade expansion valve 36, and then, the refrigerant is sent to the cascade heat exchanger 35. In the

22

cascade heat exchanger 35, the secondary-side refrigerant flowing in the secondary-side flow path 35a is evaporated into a low-pressure gas refrigerant and is sent to the secondary-side switching mechanism 22, and the primary-side refrigerant flowing in the primary-side flow path 35b of the cascade heat exchanger 35 is condensed. Then, the secondary-side low-pressure gas refrigerant sent to the first switching valve 22a of the secondary-side switching mechanism 22 is returned to the suction side of the secondary-side compressor 21 through the suction flow path 23 and the secondary-side accumulator 30.

Motion during the heating operation is performed in such a manner.

(9-3) Cooling Main Operation

During the cooling main operation, for example, the utilization-side heat exchangers 52a and 52b in the utilization units 3a and 3b function as refrigerant evaporators, and the utilization-side heat exchanger 52c in the utilization unit 3c functions as a refrigerant radiator. In the cooling main operation, the cascade heat exchanger 35 functions as a radiator for the secondary-side refrigerant. In the cooling main operation, the primary-side refrigerant circuit 5a and the secondary-side refrigerant circuit 10 of the refrigeration cycle apparatus 1 are configured as illustrated in FIG. 5. Arrows attached to the primary-side refrigerant circuit 5a and arrows attached to the secondary-side refrigerant circuit 10 in FIG. 5 indicate flows of the refrigerant during the cooling main operation.

Specifically, in the primary-side unit 5, the primary-side switching mechanism 72 is switched to the fifth connection state (the state depicted by solid lines in the primary-side switching mechanism 72 in FIG. 5) to cause the cascade heat exchanger 35 to function as an evaporator for the primary-side refrigerant. Accordingly, in the primary-side unit 5, the primary-side refrigerant discharged from the primary-side compressor 71 passes through the primary-side switching mechanism 72 and exchanges heat with outdoor air supplied from the primary-side fan 75 in the primary-side heat exchanger 74 to be condensed. The primary-side refrigerant condensed in the primary-side heat exchanger 74 passes through the primary-side first expansion valve 76 controlled into a fully opened state, and a part of the refrigerant flows toward the first liquid shutoff valve 108 through the primary-side subcooling heat exchanger 103, and another part of the refrigerant branches into the primary-side subcooling circuit 104. The refrigerant flowing in the primary-side subcooling circuit 104 is decompressed when passing through the primary-side subcooling expansion valve 104a. The refrigerant flowing from the primary-side first expansion valve 76 toward the first liquid shutoff valve 108 exchanges heat with the refrigerant decompressed by the primary-side subcooling expansion valve 104a and flowing in the primary-side subcooling circuit 104 in the primary-side subcooling heat exchanger 103, and is cooled until reaching a subcooled state. The refrigerant in the subcooled state flows through the primary-side first connection pipe 111, the second liquid shutoff valve 106, and the second refrigerant pipe 114 in that order, and is decompressed by the primary-side second expansion valve 102. At this time, for example, a valve opening degree of the primary-side second expansion valve 102 is controlled such that the degree of superheating of the refrigerant sucked into the primary-side compressor 71 satisfies a predetermined condition. When flowing in the primary-side flow path 35b of the cascade heat exchanger 35, the primary-side refrigerant decompressed by the pri-

23

mary-side second expansion valve 102 evaporates by exchanging heat with the secondary-side refrigerant in through the secondary-side flow path 35a, and flows toward the second gas shutoff valve 107 through the first refrigerant pipe 113. The refrigerant having passed through the second gas shutoff valve 107 passes through the primary-side second connection pipe 112 and the first gas shutoff valve 109, and then reaches the primary-side switching mechanism 72. The refrigerant having passed through the primary-side switching mechanism 72 joins the refrigerant having flowed through the primary-side subcooling circuit 104, and is then sucked into the primary-side compressor 71 via the primary-side accumulator 105.

In the cascade unit 2, the secondary-side switching mechanism 22 is switched to the third connection state in which the discharge flow path 24 and the third pipe 25 are connected by the first switching valve 22a and the discharge flow path 24 and the first pipe 28 are connected by the second switching valve 22b to cause the cascade heat exchanger 35 to function as a radiator for the secondary-side refrigerant. The opening degree of the cascade expansion valve 36 is adjusted. In the first to third branch units 6a, 6b, and 6c, the first regulating valve 66c and the second regulating valves 67a and 67b are controlled to the opened state, and the first regulating valves 66a and 66b and the second regulating valve 67c are controlled to the closed state. Accordingly, the utilization-side heat exchangers 52a and 52b in the utilization units 3a and 3b function as refrigerant evaporators, and the utilization-side heat exchanger 52c in the utilization unit 3c functions as a refrigerant radiator. The utilization-side heat exchangers 52a and 52b in the utilization units 3a and 3b and the suction side of the secondary-side compressor 21 in the cascade unit 2 are connected via the secondary-side second connection pipe 9, and the utilization-side heat exchanger 52c in the utilization unit 3c and the discharge side of the secondary-side compressor 21 in the cascade unit 2 are connected via the secondary-side first connection pipe 8. In addition, an opening degree of the secondary-side subcooling expansion valve 48a is controlled such that a degree of subcooling of the secondary-side refrigerant flowing through the outlet of the secondary-side subcooling heat exchanger 47 toward the secondary-side third connection pipe 7 satisfies a predetermined condition. The bypass expansion valve 46a is controlled to the closed state. In the utilization units 3a, 3b, and 3c, the opening degrees of the utilization-side expansion valves 51a, 51b, and 51c are adjusted.

In the cooling main operation, the secondary-side refrigerant circuit 10 controls capacity, for example, by controlling the frequency of the secondary-side compressor 21 such that evaporation temperature in a heat exchanger functioning as an evaporator for the secondary-side refrigerant between the utilization-side heat exchanger 52a, 52b, and 52c becomes a predetermined secondary-side evaporation target temperature. The opening degree of the cascade expansion valve 36 is adjusted such that the secondary-side refrigerant flowing in the cascade heat exchanger 35 has a critical pressure or less. The primary-side refrigerant circuit 5a controls capacity, for example, by controlling a frequency of the primary-side compressor 71 such that evaporation temperature of the primary-side refrigerant in the primary-side flow path 35b of the cascade heat exchanger 35 becomes a predetermined primary-side evaporation target temperature. In such a manner, in the cooling operation, either or both of the control for increasing the valve opening degree of the cascade expansion valve 36 and the control for increasing the frequency of the primary-side compressor 71 in the

24

primary-side refrigerant circuit 5a are executed, and thus, the carbon dioxide refrigerant flowing in the cascade heat exchanger 35 is controlled so as not to exceed a critical point.

In such a secondary-side refrigerant circuit 10, a part of the secondary-side high-pressure refrigerant compressed and discharged by the secondary-side compressor 21 is sent to the secondary-side first connection pipe 8 through the second switching valve 22b of the secondary-side switching mechanism 22, the first pipe 28, and the first shutoff valve 32, and the rest is sent to the secondary-side flow path 35a of the cascade heat exchanger 35 through the first switching valve 22a of the secondary-side switching mechanism 22 and the third pipe 25.

Then, the high-pressure refrigerant sent to the secondary-side first connection pipe 8 is sent to the first branch pipe 63c. The high-pressure refrigerant sent to the first branch pipe 63c is sent to the utilization-side heat exchanger 52c in the utilization unit 3c through the first regulating valve 66c and the junction pipe 62c.

Then, the high-pressure refrigerant sent to the utilization-side heat exchanger 52c exchanges heat with indoor air supplied by the indoor fan 53c in the utilization-side heat exchanger 52c. The refrigerant flowing in the utilization-side heat exchanger 52c thus radiates heat. The indoor air is heated and is supplied into the indoor space, and the utilization unit 3c performs the heating operation. The refrigerant having radiated heat in the utilization-side heat exchanger 52c flows in the second utilization pipe 56c, and the flow rate of the refrigerant is adjusted at the utilization-side expansion valve 51c. After that, the refrigerant having flowed through the second connecting tube 16c is sent to the third branch pipe 61c in the branch unit 6c.

Then, the refrigerant sent to the third branch pipe 61c is sent to the secondary-side third connection pipe 7.

The high-pressure refrigerant sent to the secondary-side flow path 35a of the cascade heat exchanger 35 exchanges heat with the primary-side refrigerant flowing in the primary-side flow path 35b in the cascade heat exchanger 35 to radiate heat. The flow rate of the secondary-side refrigerant having radiated heat in the cascade heat exchanger 35 is adjusted at the cascade expansion valve 36, and then the secondary-side refrigerant flows into the secondary-side receiver 45. A part of the refrigerant having flowed out of the secondary-side receiver 45 branches into the secondary-side subcooling circuit 48, is decompressed at the secondary-side subcooling expansion valve 48a, and then joins into the suction flow path 23. In the secondary-side subcooling heat exchanger 47, another part of the refrigerant having flowed out of the secondary-side receiver 45 is cooled by the refrigerant flowing in the secondary-side subcooling circuit 48, is then sent to the secondary-side third connection pipe 7 through the third shutoff valve 31, and joins the refrigerant having radiated heat in the utilization-side heat exchanger 52c.

Then, the refrigerant having joined in the secondary-side third connection pipe 7 is branched into two portions to be sent to the third branch pipes 61a and 61b of the branch units 6a and 6b. Thereafter, the refrigerant having flowed in the second connecting tubes 16a and 16b is sent to the second utilization pipes 56a and 56b of the first and second utilization units 3a and 3b. The refrigerant flowing in the second utilization pipes 56a and 56b passes through the utilization-side expansion valves 51a and 51b in the utilization units 3a and 3b.

The refrigerant having passed through the utilization-side expansion valves 51a and 51b whose opening degrees are

25

adjusted exchanges heat with indoor air supplied by the indoor fans **53a** and **53b** in the utilization-side heat exchangers **52a** and **52b**. The refrigerant flowing in the utilization-side heat exchangers **52a** and **52b** is thus evaporated into a low-pressure gas refrigerant. The indoor air is cooled and is supplied into the indoor space. The indoor space is thus cooled. The low-pressure gas refrigerant evaporated in the utilization-side heat exchangers **52a** and **52b** is sent to the junction pipes **62a** and **62b** of the first and second branch units **6a** and **6b**.

The low-pressure gas refrigerant sent to the junction pipes **62a** and **62b** is sent to the secondary-side second connection pipe **9** via the second regulating valves **67a** and **67b** and the second branch pipes **64a** and **64b**, to join.

The low-pressure gas refrigerant sent to the secondary-side second connection pipe **9** is returned to the suction side of the secondary-side compressor **21** via the second shutoff valve **33**, the second pipe **29**, the suction flow path **23**, and the secondary-side accumulator **30**.

Motion during the cooling main operation is performed in such a manner.

(9-4) Heating Main Operation

During the heating main operation, for example, the utilization-side heat exchangers **52a** and **52b** in the utilization units **3a** and **3b** function as refrigerant radiators, and the utilization-side heat exchanger **52c** functions as a refrigerant evaporator. In the heating main operation, the cascade heat exchanger **35** functions as an evaporator for the secondary-side refrigerant. In the heating main operation, the primary-side refrigerant circuit **5a** and the secondary-side refrigerant circuit **10** of the refrigeration cycle apparatus **1** are configured as illustrated in FIG. 6. Arrows attached to the primary-side refrigerant circuit **5a** and arrows attached to the secondary-side refrigerant circuit **10** in FIG. 6 indicate flows of the refrigerant during the heating main operation.

Specifically, in the primary-side unit **5**, the primary-side switching mechanism **72** is switched to a sixth operating state to cause the cascade heat exchanger **35** to function as a radiator for the primary-side refrigerant. The sixth operating state of the primary-side switching mechanism **72** corresponds to a connection state depicted by broken lines in the primary-side switching mechanism **72** in FIG. 6. Accordingly, in the primary-side unit **5**, the primary-side refrigerant discharged from the primary-side compressor **71**, having passed through the primary-side switching mechanism **72** and the first gas shutoff valve **109** passes through the primary-side second connection pipe **112** and the second gas shutoff valve **107** and is sent to the primary-side flow path **35b** of the cascade heat exchanger **35**. The refrigerant flowing in the primary-side flow path **35b** of the cascade heat exchanger **35** is condensed by exchanging heat with the secondary-side refrigerant flowing in the secondary-side flow path **35a**. When flowing in the second refrigerant pipe **114**, the primary-side refrigerant condensed in the cascade heat exchanger **35** passes through the primary-side second expansion valve **102** controlled to the fully opened state. Then, the primary-side refrigerant flows through the second liquid shutoff valve **106**, the primary-side first connection pipe **111**, the first liquid shutoff valve **108**, and the primary-side subcooling heat exchanger **103** in that order, and is decompressed by the primary-side first expansion valve **76**. During the heating main operation, the primary-side subcooling expansion valve **104a** is controlled to the closed state. Accordingly, the refrigerant does not flow into the primary-side subcooling circuit **104** and does not exchange

26

heat in the primary-side subcooling heat exchanger **103**. The valve opening degree of the primary-side first expansion valve **76** is controlled such that, for example, the degree of superheating of the refrigerant sucked into the primary-side compressor **71** satisfies a predetermined condition. The refrigerant decompressed by the primary-side first expansion valve **76** evaporates by exchanging heat with outdoor air supplied from the primary-side fan **75** in the primary-side heat exchanger **74**, passes through the primary-side switching mechanism **72** and the primary-side accumulator **105**, and is sucked into the primary-side compressor **71**.

In the cascade unit **2**, the secondary-side switching mechanism **22** is switched to the second connection state. In the second connection state of the secondary-side switching mechanism **22**, the discharge flow path **24** and the first pipe **28** are connected by the second switching valve **22b**, and the third pipe **25** and the suction flow path **23** are connected by the first switching valve **22a**. The cascade heat exchanger **35** thus functions as an evaporator for the secondary-side refrigerant. The opening degree of the cascade expansion valve **36** is adjusted. In the first to third branch units **6a**, **6b**, and **6c**, the first regulating valves **66a** and **66b** and the second regulating valve **67c** are controlled to the opened state, and the first regulating valve **66c** and the second regulating valves **67a** and **67b** are controlled to the closed state. Accordingly, the utilization-side heat exchangers **52a** and **52b** in the utilization units **3a** and **3b** function as refrigerant radiators, and the utilization-side heat exchanger **52c** in the utilization unit **3c** functions as a refrigerant evaporator. The utilization-side heat exchanger **52c** in the utilization unit **3c** and the suction side of the secondary-side compressor **21** in the cascade unit **2** are connected via the first utilization pipe **57c**, the first connecting tube **15c**, the junction pipe **62c**, the second branch pipe **64c**, and the secondary-side second connection pipe **9**. The utilization-side heat exchangers **52a** and **52b** in the utilization units **3a** and **3b** and the discharge side of the secondary-side compressor **21** in the cascade unit **2** are connected via the discharge flow path **24**, the first pipe **28**, the secondary-side first connection pipe **8**, the first branch pipes **63a** and **63b**, the junction pipes **62a** and **62b**, the first connecting tubes **15a** and **15b**, and the first utilization pipes **57a** and **57b**. The secondary-side subcooling expansion valve **48a** and the bypass expansion valve **46a** are controlled to the closed state. In the utilization units **3a**, **3b**, and **3c**, the opening degrees of the utilization-side expansion valves **51a**, **51b**, and **51c** are adjusted.

In the heating main operation, the secondary-side refrigerant circuit **10** controls capacity, for example, by controlling the frequency of the secondary-side compressor **21** so as to process a load in a heat exchanger functioning as a radiator for the secondary-side refrigerant between the utilization-side heat exchangers **52a**, **52b**, and **52c**. As a result, in the heating main operation, the secondary-side refrigerant discharged from the secondary-side compressor **21** is controlled to be in the critical state exceeding the critical pressure. The primary-side refrigerant circuit **5a** controls capacity, for example, by controlling the frequency of the primary-side compressor **71** such that condensation temperature of the primary-side refrigerant in the primary-side flow path **35b** of the cascade heat exchanger **35** becomes a predetermined primary-side condensation target temperature.

In such a secondary-side refrigerant circuit **10**, the secondary-side high-pressure refrigerant compressed and discharged by the secondary-side compressor **21** is sent to the secondary-side first connection pipe **8** through the second

27

switching valve 22b of the secondary-side switching mechanism 22, the first pipe 28, and the first shutoff valve 32.

The high-pressure refrigerant sent to the secondary-side first connection pipe 8 is branched into two portions to be sent to the first branch pipes 63a and 63b of the first branch unit 6a and the second branch unit 6b connected to the first utilization unit 3a and the second utilization unit 3b which are utilization units in operation. The high-pressure refrigerant sent to the first branch pipes 63a and 63b is sent to the utilization-side heat exchangers 52a and 52b in the first utilization unit 3a and the second utilization unit 3b via the first regulating valves 66a and 66b, the junction pipes 62a and 62b, and the first connecting tubes 15a and 15b.

The high-pressure refrigerant sent to the utilization-side heat exchangers 52a and 52b exchanges heat with indoor air supplied by the indoor fans 53a and 53b in the utilization-side heat exchangers 52a and 52b. The refrigerant flowing in the utilization-side heat exchangers 52a and 52b thus radiates heat. The indoor air is heated and supplied into the indoor space. The indoor space is thus heated. The refrigerant having radiated heat in the utilization-side heat exchangers 52a and 52b flows in the second utilization pipes 56a and 56b, and passes through the utilization-side expansion valves 51a and 51b whose opening degree is adjusted. The secondary-side refrigerant that has passed through the utilization-side expansion valves 51a and 51b has the critical pressure or less. Thereafter, the refrigerant having flowed through the second connecting tubes 16a and 16b is sent to the secondary-side third connection pipe 7 via the third branch pipes 61a and 61b of the branch units 6a and 6b.

A part of the refrigerant sent to the secondary-side third connection pipe 7 is sent to the third branch pipe 61c of the branch unit 6c, and the rest flows toward the third shutoff valve 31.

Then, the refrigerant sent to the third branch pipe 61c flows in the second utilization pipe 56c of the utilization unit 3c via the second connecting tube 16c, and is sent to the utilization-side expansion valve 51c.

The refrigerant having passed through the utilization-side expansion valve 51c whose opening degree is adjusted exchanges heat with indoor air supplied by the indoor fan 53c in the utilization-side heat exchanger 52c. The refrigerant flowing in the utilization-side heat exchanger 52c is thus evaporated into a low-pressure gas refrigerant. The indoor air is cooled and is supplied into the indoor space. The indoor space is thus cooled. The low-pressure gas refrigerant evaporated in the utilization-side heat exchanger 52c passes through the first utilization pipe 57c and the first connecting tube 15c to be sent to the junction pipe 62c.

The low-pressure gas refrigerant sent to the junction pipe 62c is sent to the secondary-side second connection pipe 9 through the second regulating valve 67c and the second branch pipe 64c.

The low-pressure gas refrigerant sent to the secondary-side second connection pipe 9 is returned to the suction side of the secondary-side compressor 21 via the second shutoff valve 33, the second pipe 29, the suction flow path 23, and the secondary-side accumulator 30.

The refrigerant flowing toward the third shutoff valve 31 is sent to the cascade expansion valve 36. The refrigerant sent to the cascade expansion valve 36 passes through the cascade expansion valve 36 whose opening degree is adjusted, and then exchanges heat with the primary-side refrigerant flowing in the primary-side flow path 35b in the secondary-side flow path 35a of the cascade heat exchanger 35. As a result, the refrigerant flowing in the secondary-side flow path 35a of the cascade heat exchanger 35 evaporates

28

to become a low-pressure gas refrigerant, and is sent to the first switching valve 22a of the secondary-side switching mechanism 22. The low-pressure gas refrigerant sent to the first switching valve 22a of the secondary-side switching mechanism 22 joins the low-pressure gas refrigerant evaporated in the utilization-side heat exchanger 52c in the suction flow path 23. The refrigerant thus joined is returned to the suction side of the secondary-side compressor 21 via the secondary-side accumulator 30.

Motion during the heating main operation is performed in such a manner.

(9-5) Oil Return Operation

When a predetermined oil return condition is satisfied during the heating operation and the heating main operation, the operation is performed with the connection state of the secondary-side refrigerant circuit 10 temporarily set as the connection state during the cooling operation to perform the oil return operation. As a result, the refrigerating machine oil retained at a lower end of the vessel body 90 of the secondary-side receiver 45 can be extracted from the secondary-side receiver 45 through the second refrigerant pipe 97 of the fifth pipe 27 extending from a lower end of the secondary-side receiver 45.

The predetermined oil return condition is not limited, and for example, the heating operation and the heating main operation may be performed for a predetermined time. An end of the oil return operation is not limited, and for example, the oil return operation may be performed for a predetermined time.

(10) Secondary-Side Receiver

FIG. 7 is a schematic configuration diagram of the secondary-side receiver 45.

The secondary-side receiver 45 includes the vessel body 90, a first refrigerant pipe 96, a second refrigerant pipe 97, and a third refrigerant pipe 98.

The vessel body 90 is a substantially cylindrical vessel having an internal volume corresponding to the amount of refrigerant filled in the secondary-side refrigerant circuit 10, and temporarily reserves the refrigerant flowing in the secondary-side refrigerant circuit 10. The vessel body 90 includes an upper part 91, a lower peripheral surface part 92, and a bottom part 93, which are welded and fixed to each other. The upper part 91 has a cylindrical shape in which an internal space is formed so as to open downward. The lower peripheral surface part 92 is disposed under the upper part 91 and has a cylindrical shape penetrating in an up-down direction. The bottom part 93 is a columnar member whose thickness direction is the up-down direction. The bottom part 93 has an upper surface 93a facing upward, toward a space inside the secondary-side receiver 45, and a lower surface 93b facing downward, toward a space outside the secondary-side receiver 45. In the present embodiment, both the upper surface 93a and the lower surface 93b extend in a horizontal plane. The bottom part 93 has a portion penetrating in the up-down direction at a center in plan view, and an end of the second refrigerant pipe 97 on a side connected to the secondary-side receiver 45 is inserted from below. The bottom part 93 and the second refrigerant pipe 97 are welded to each other.

In the present embodiment, the first refrigerant pipe 96 is, for example, a pipe extending laterally from a part of a peripheral surface of the vessel body 90, and constitutes a part of the fourth pipe 26 in the secondary-side refrigerant

29

circuit 10. The first refrigerant pipe 96 is welded to the upper part 91 of the vessel body 90. The first refrigerant pipe 96 has a portion extending downward inside the secondary-side receiver 45, and an end located inside the secondary-side receiver 45 has an opening 96a formed to be inclined with respect to an axial direction in which the pipe extends. The end of the first refrigerant pipe 96 in the secondary-side receiver 45 is in contact with the upper surface 93a of the bottom part 93 or extends to a position in front of the upper surface 93a of the bottom part 93.

The second refrigerant pipe 97 is a pipe extending downward from the bottom part 93 of the vessel body 90, and constitutes a part of the fifth pipe 27 in the secondary-side refrigerant circuit 10. The second refrigerant pipe 97 is welded to the bottom part 93 in a state of being inserted into a penetrating portion in the up-down direction formed in the bottom part 93. In the present embodiment, the upper end 97a of the second refrigerant pipe 97 is disposed at a height position equal to or lower than a height position of a portion of the upper surface 93a of the bottom part 93 where the second refrigerant pipe 97 is connected. Specifically, in the present embodiment, the upper end 97a of the second refrigerant pipe 97 is disposed on the same horizontal plane as the upper surface 93a of the bottom part 93.

The third refrigerant pipe 98 is a pipe extending laterally from a part of the peripheral surface of the vessel body 90, and constitutes a part of the bypass circuit 46 in the secondary-side refrigerant circuit 10. The third refrigerant pipe 98 is welded to the upper part 91 of the vessel body 90.

(11) Characteristics of Embodiment

In the secondary-side refrigerant circuit 10 of the refrigeration cycle apparatus 1 according to the present embodiment, the carbon dioxide refrigerant and the PAG oil which are incompatible with each other are used. Here, the PAG oil has a higher density than the carbon dioxide refrigerant in a use condition in the secondary-side refrigerant circuit 10. Therefore, in the secondary-side receiver 45, the PAG oil is likely to be located below the carbon dioxide refrigerant.

In the secondary-side receiver 45 according to the present embodiment, the second refrigerant pipe 97 is provided so as to penetrate the bottom part 93 in the up-down direction. The upper end 97a of the second refrigerant pipe 97 is disposed at the same height position as the upper surface 93a of the bottom part 93 or at a height position lower than the upper surface 93a of the bottom part 93. As a result, the PAG oil flowing into the secondary-side receiver 45 via the first refrigerant pipe 96 and located below carbon dioxide refrigerant passes through the second refrigerant pipe 97 and is efficiently sent out of the secondary-side receiver 45. Therefore, the refrigerating machine oil is prevented from being continuously reserved in the secondary-side receiver 45.

In the refrigeration cycle apparatus 1 according to the present embodiment, the carbon dioxide refrigerant discharged from the secondary-side compressor 21 of the secondary-side refrigerant circuit 10 is not in the supercritical state by exchanging heat with the primary-side refrigerant flowing in the primary-side refrigerant circuit 5a in the cascade heat exchanger 35 or by controlling the valve opening degree of the cascade expansion valve 36, and is sent to the secondary-side receiver 45. Therefore, even if the carbon dioxide refrigerant and the refrigerating machine oil are phase-separated in the secondary-side receiver 45, it is possible to prevent the refrigerating machine oil from con-

30

tinuously accumulating in the secondary-side receiver 45 while avoiding the supercritical state in which the behavior tends to be unstable.

(12) Other Embodiments

(12-1) Another Embodiment A

In the above embodiment, as an example, a case has been described where the upper surface 93a of the bottom part 93 of the secondary-side receiver 45 is a horizontal plane.

Alternatively, for example, as illustrated in FIG. 8, the bottom part 93 of the secondary-side receiver 45 may have an upper curved surface 193a that is curved so as to gently protrude downward and constitutes an inner bottom surface of the secondary-side receiver 45, instead of the upper surface 93a according to the above embodiment. The upper curved surface 193a has a curved shape that gradually descends toward the upper end 97a of the second refrigerant pipe 97. The upper end 97a of the second refrigerant pipe 97 is located at a lowermost end of the upper curved surface 193a.

In this case, since the refrigerating machine oil located on the upper curved surface 193a is guided to the upper end 97a of the second refrigerant pipe 97 by a weight of the refrigerating machine oil, the refrigerating machine oil can be efficiently discharged from the secondary-side receiver 45.

(12-2) Another Embodiment B

In the another embodiment A, as an example, a case has been described where the bottom part 93 of the secondary-side receiver 45 has the upper curved surface 193a.

Alternatively, for example, as illustrated in FIG. 9, the bottom part 93 of the secondary-side receiver 45 may have a lower curved surface 193b that is curved so as to protrude downward similarly to the upper curved surface 193a and constitutes an outer bottom surface of the secondary-side receiver 45, instead of the lower surface 93b.

In this case, not only the effects of the above embodiment and the another embodiment A can be obtained, but also the thickness of the bottom part 93 in the up-down direction can be made constant, and a pressure resistance strength of the secondary-side receiver 45 can be easily secured.

(12-3) Another Embodiment C

In the above embodiment, as an example, a case has been described where the height position of the upper end 97a of the second refrigerant pipe 97 is equal to or lower than the height position of the portion of the upper surface 93a of the bottom part 93 where the second refrigerant pipe 97 is connected.

However, the height position of the upper end 97a of the second refrigerant pipe 97 is not limited to the height position as described above.

For example, as illustrated in FIG. 10, the upper end 197a of the second refrigerant pipe 197 may be located slightly above a portion of the upper surface 93a of the bottom part 93 where the second refrigerant pipe 197 is connected. For example, the height position of the upper end 197a of the second refrigerant pipe 197 may be higher than the portion of the upper surface 93a of the bottom part 93 where the second refrigerant pipe 197 is connected, and may be lower than a height position 15 mm higher than the portion. The height position of the upper end 197a of the second refrigerant pipe 197 may be equal to or lower than a height

31

position 10 mm higher than the portion of the upper surface **93a** of the bottom part **93** where the second refrigerant pipe **197** is connected. A difference in a height direction between the upper end **197a** of the second refrigerant pipe **197** and the portion of the upper surface **93a** of the bottom part **93** where the second refrigerant pipe **197** is connected may be $\frac{1}{100}$ or less of a height of the vessel body **90** of the secondary-side receiver **45**. Since the upper end **197a** of the second refrigerant pipe **197** protrudes further upward than the upper surface **93a** of the bottom part **93** as described above, a pipe protruding margin can be secured to facilitate manufacturing.

A portion above the upper surface **93a** of the bottom part **93** near the upper end **197a** of the second refrigerant pipe **197** may be flared such that a flow path area increases upward.

(12-4) Another Embodiment D

In the above embodiment, as an example, a case has been described where the first refrigerant pipe **96** is provided so as to extend laterally from a part of a peripheral surface of the secondary-side receiver **45**.

However, a connection form of the first refrigerant pipe **96** with the secondary-side receiver **45** is not limited to the above case.

For example, as illustrated in FIG. **11**, the first refrigerant pipe **196** may be provided so as to extend downward from the bottom part **93** of the secondary-side receiver **45**. In this case, the upper end **196a** of the first refrigerant pipe **196** may be configured similarly to the upper end **97a** of the second refrigerant pipe **97** according to the above embodiment, or similarly to the upper end **197a** of the second refrigerant pipe **197** of the another embodiment C.

(12-5) Another Embodiment E

In the above embodiment, as an example, a case has been described where the second refrigerant pipe **97** is provided so as to penetrate the bottom part **93** in the up-down direction in the secondary-side receiver **45** as a high-pressure receiver provided at a portion in which the high-pressure refrigerant flows in the secondary-side refrigerant circuit **10**.

Alternatively, for example, as an intermediate-pressure receiver **18** in a refrigerant circuit **210** of an air conditioner **201** illustrated in FIG. **12**, a second refrigerant pipe **19** may be provided so as to penetrate the bottom part in the up-down direction.

The air conditioner **201** is a device that performs a vapor compression refrigeration cycle to condition air in a target space. The air conditioner **201** mainly includes an outdoor unit **202**, an indoor unit **203**, a liquid-side connection pipe **207** and a gas-side connection pipe **208** that connect the outdoor unit **202** and the indoor unit **203**, and a controller **4** that controls motion of the air conditioner **201**. The air conditioner **201** performs a refrigeration cycle in which a refrigerant sealed in the refrigerant circuit **210** is compressed, radiates heat, is decompressed, evaporated, and then compressed again. The air conditioner **201** is filled with a refrigerant and a refrigerating machine oil that is incompatible with the refrigerant and has a higher density than the refrigerant. Specifically, the refrigerant is a carbon dioxide refrigerant, and the refrigerating machine oil is PAG oil.

The outdoor unit **202** is connected to the indoor unit **203** via the liquid-side connection pipe **207** and the gas-side connection pipe **208**, and constitutes a part of the refrigerant

32

circuit **210**. The outdoor unit **202** mainly includes a compressor **221**, a four-way switching valve **222**, an outdoor heat exchanger **235**, a first refrigerant pipe **17**, a first outdoor expansion valve **17a**, the intermediate-pressure receiver **18**, the second refrigerant pipe **19**, a second outdoor expansion valve **19a**, an outdoor fan **11**, a liquid-side shutoff valve **231**, and a gas-side shutoff valve **232**.

The compressor **221** is a device that compresses a low-pressure refrigerant in the refrigeration cycle to a high pressure. The compressor **221** used herein is a closed compressor in which a rotary type, scroll type, or other positive-displacement compression element is driven to rotate by a compressor motor. The compressor motor is for changing a capacity, and an operating frequency can be controlled by an inverter.

By switching the connection state, the four-way switching valve **222** can switch between a connection state in which a cooling operation is performed by connecting a suction side of the compressor **221** and the gas-side shutoff valve **232** while connecting a discharge side of the compressor **221** and the outdoor heat exchanger **235**, and a connection state in which a heating operation is performed by connecting the suction side of the compressor **221** and the outdoor heat exchanger **235** while connecting the discharge side of the compressor **221** and the gas-side shutoff valve **232**.

The outdoor heat exchanger **235** functions as a radiator for the high-pressure refrigerant in the refrigeration cycle during the cooling operation, and functions as an evaporator for the low-pressure refrigerant in the refrigeration cycle during the heating operation. The outdoor heat exchanger **235** includes a plurality of heat transfer fins and a plurality of heat transfer tubes penetrated through and fixed to the plurality of heat transfer fins. The outdoor heat exchanger **235** is an air heat exchanger that causes heat exchange between the refrigerant and air flowing outside.

The outdoor fan **11** sucks outdoor air into the outdoor unit **202**, causes the outdoor air to exchange heat with the refrigerant in the outdoor heat exchanger **235**, and then generates an air flow to be discharged to the outside. The outdoor fan **11** is driven to rotate by an outdoor fan motor.

The first refrigerant pipe **17** connects an end of the outdoor heat exchanger **235** on an opposite side of the compressor **221** and the intermediate-pressure receiver **18**. The first refrigerant pipe **17** is provided with a first outdoor expansion valve **17a** whose valve opening degree is controllable. During the cooling operation, the opening degree of the first outdoor expansion valve **17a** is controlled in order to send the intermediate-pressure refrigerant obtained by decompressing the high-pressure refrigerant to the intermediate-pressure receiver **18**. During the heating operation, the opening degree of the first outdoor expansion valve **17a** is controlled in order to send the low-pressure refrigerant obtained by decompressing the intermediate-pressure refrigerant to the outdoor heat exchanger **235**.

The intermediate-pressure receiver **18** is provided between the first outdoor expansion valve **17a** and the second outdoor expansion valve **19a** in the refrigerant circuit **210**. Note that the details of the specific structure of the intermediate-pressure receiver **18** are similar to those of the secondary-side receiver **45** according to the above embodiment.

The second refrigerant pipe **19** connects the bottom part of the intermediate-pressure receiver **18** and the liquid-side shutoff valve **231**. The second refrigerant pipe **19** is provided with a second outdoor expansion valve **19a** whose valve opening degree is controllable. During the heating operation, the opening degree of the second outdoor expansion valve

19a is controlled in order to send the intermediate-pressure refrigerant obtained by decompressing the high-pressure refrigerant to the intermediate-pressure receiver 18. During the cooling operation, the opening degree of the second outdoor expansion valve 19a is controlled in order to send the low-pressure refrigerant obtained by decompressing the intermediate-pressure refrigerant to the indoor heat exchanger 252.

The liquid-side shutoff valve 231 is a manual valve disposed at a connecting part of the outdoor unit 202 with the liquid-side connection pipe 207. The gas-side shutoff valve 232 is a manual valve disposed at a connecting part of the outdoor unit 202 with the gas-side connection pipe 208.

The outdoor unit 202 includes an outdoor unit control unit 4a that controls motion of each element constituting the outdoor unit 202. The indoor unit 203 is installed on, for example, an indoor wall surface as a target space. The indoor unit 203 is connected to the outdoor unit 202 via the liquid-side connection pipe 207 and the gas-side connection pipe 208, and constitutes a part of the refrigerant circuit 210. The indoor unit 203 includes the indoor heat exchanger 252, an indoor fan 253, and the like.

The indoor heat exchanger 252 has a liquid side connected to the liquid-side connection pipe 207 and a gas side end connected to the gas-side connection pipe 208. The indoor heat exchanger 252 functions as an evaporator for the low-pressure refrigerant in the refrigeration cycle during the cooling operation, and functions as a condenser for the high-pressure refrigerant in the refrigeration cycle during the heating operation. The indoor heat exchanger 252 includes a plurality of heat transfer fins and a plurality of heat transfer tubes penetrated through and fixed to the plurality of heat transfer fins.

The indoor fan 253 sucks indoor air into the indoor unit 203, causes the indoor air to exchange heat with the refrigerant in the indoor heat exchanger 252, and then generates an air flow to be discharged to the outside.

The indoor unit 203 includes an indoor unit control unit 4b that controls motion of each element constituting the indoor unit 203.

As in the above embodiment, the intermediate-pressure receiver 18 described above can efficiently discharge the refrigerating machine oil which is incompatible with the carbon dioxide refrigerant, has a higher density than the carbon dioxide refrigerant, and tends to be located below the carbon dioxide refrigerant through the second refrigerant pipe 19.

(12-6) Another Embodiment F

In the above embodiment, description has been made by exemplifying the refrigeration cycle apparatus 1 in which one cascade unit 2 is connected to one primary-side unit 5.

Alternatively, as illustrated in FIG. 13, for example, by connecting are a plurality of cascade units, namely, a first cascade unit 2a, a second cascade unit 2b, and a third cascade unit 2c, in parallel to each other to one primary-side unit 5, the refrigeration cycle apparatus 1 may include a first secondary-side refrigerant circuit 10a including a first cascade circuit 12a, a second secondary-side refrigerant circuit 10b including a second cascade circuit 12b, and a third secondary-side refrigerant circuit 10c including a third cascade circuit 12c. Note that, in FIG. 13, an internal structure of each of the first cascade unit 2a, the second cascade unit 2b, and the third cascade unit 2c is similar to that of the cascade unit 2 according to the above embodiment, and thus only a part of each cascade unit is illustrated.

Although not illustrated, each of the first cascade unit 2a, the second cascade unit 2b, and the third cascade unit 2c is connected to the plurality of branch units 6a, 6b, and 6c and the plurality of utilization units 3a, 3b, and 3c as in the above embodiment. Specifically, the first cascade unit 2a is connected to a plurality of branch units and utilization units via a secondary-side third connection pipe 7a, a secondary-side first connection pipe 8a, and a secondary-side second connection pipe 9a. The second cascade unit 2b is connected, via a secondary-side third connection pipe 7b, a secondary-side first connection pipe 8b, and a secondary-side second connection pipe 9b, to a plurality of branch units and utilization units different from those connected to the first cascade unit 2a. The third cascade unit 2c is connected, via a secondary-side third connection pipe 7c, a secondary-side first connection pipe 8c, and a secondary-side second connection pipe 9c, to another plurality of branch units and utilization units different from those connected to the first cascade unit 2a and different from those connected to the second cascade unit 2b.

Here, the primary-side unit 5 and the first cascade unit 2a are connected via a primary-side first connection pipe 111a and a primary-side second connection pipe 112a. The primary-side unit 5 and the second cascade unit 2b are connected via a primary-side first connection pipe 111b branched from the primary-side first connection pipe 111a and a primary-side second connection pipe 112b branched from the primary-side second connection pipe 112a. The primary-side unit 5 and the third cascade unit 2c are connected via a primary-side first connection pipe 111c branched from the primary-side first connection pipe 111a and a primary-side second connection pipe 112c branched from the primary-side second connection pipe 112a.

Here, each of the first cascade unit 2a, the second cascade unit 2b, and the third cascade unit 2c includes a primary-side second expansion valve 102 whose opening degree is controlled by the first cascade unit 2a, the second cascade unit 2b, and the third cascade unit 2c. Furthermore, a first cascade-side control unit 20a included in the first cascade unit 2a, a second cascade-side control unit 20b included in the second cascade unit 2b, and a third cascade-side control unit 20c included in the third cascade unit 2c control the opening degree of the corresponding primary-side second expansion valve 102. Similarly to the above embodiment, each of the first cascade-side control unit 20a, the second cascade-side control unit 20b, and the third cascade-side control unit 20c controls the valve opening degree of the corresponding primary-side second expansion valve 102 on the basis of conditions of the first cascade circuit 12a, the second cascade circuit 12b, and the third cascade circuit 12c controlled by the first cascade-side control unit 20a, the second cascade-side control unit 20b, and the third cascade-side control unit 20c. As a result, the primary-side refrigerant flowing through the primary-side refrigerant circuit 5a is controlled to have a flow rate of the primary-side refrigerant in the primary-side first connection pipe 111a and the primary-side second connection pipe 112a, a flow rate of the primary-side refrigerant in the primary-side first connection pipe 111b and the primary-side second connection pipe 112b, and a flow rate of the primary-side refrigerant in the primary-side first connection pipe 111c and the primary-side second connection pipe 112c so as to correspond to a difference in loads in the first secondary-side refrigerant circuit 10a, the second secondary-side refrigerant circuit 10b, and the third secondary-side refrigerant circuit 10c.

(12-7) Another Embodiment G

In the above embodiment, R32 or R410A is exemplified as the refrigerant used in the primary-side refrigerant circuit

35

5a, and carbon dioxide is exemplified as the refrigerant used in the secondary-side refrigerant circuit 10.

Alternatively, the refrigerant provided in the primary-side refrigerant circuit 5a should not be limited, and examples of the refrigerant include HFC-32, an HFO refrigerant, a refrigerant obtained by mixing HFC-32 and the HFO refrigerant, carbon dioxide, ammonia, and propane.

Furthermore, instead of the primary-side refrigerant circuit 5a in which the refrigerant flows, a heat medium circuit in which a heat medium such as water or brine flows may be used. In this case, the heat medium circuit may include a heat source that functions as a heat source or a cold source, and a pump for circulating the heat medium. In this case, the flow rate can be adjusted by the pump, and the amount of heat can be controlled by the heat source or the cold source.

The refrigerant provided in the secondary-side refrigerant circuit 10 should not be limited, and examples of the refrigerant include HFC-32, an HFO refrigerant, a refrigerant obtained by mixing HFC-32 and the HFO refrigerant, carbon dioxide, ammonia, and propane.

Note that examples of the HFO refrigerant include HFO-1234yf and HFO-1234ze.

The same refrigerant or different refrigerants may be used in the primary-side refrigerant circuit 5a and the secondary-side refrigerant circuit 10. Preferably, the refrigerant used in the secondary-side refrigerant circuit 10 has at least one of lower global warming potential (GWP), lower ozone depletion potential (ODP), lower flammability, or lower toxicity than the refrigerant used in the primary-side refrigerant circuit 5a. Here, the flammability can be compared in accordance with classifications related to ASHRAE 34 flammability, for example. Note that the toxicity can be compared, for example, in accordance with classifications related to ASHRAE 34 safety grade. In particular, when an overall content volume of the secondary-side refrigerant circuit 10 is larger than an overall content volume of the primary-side refrigerant circuit 5a, by using the refrigerant lower than the refrigerant in the primary-side refrigerant circuit 5a in at least one of the global warming potential (GWP), the ozone depletion potential (ODP), the flammability, or the toxicity in the secondary-side refrigerant circuit 10, adverse effects when a leak occurs can be reduced.

(12-8) Others

In some embodiments, the first refrigerant pipe may introduce the refrigerant into the vessel body, and the second refrigerant pipe may derive the refrigerant from the vessel body.

That the refrigerating machine oil is incompatible with the refrigerant means that the refrigerant and the refrigerating machine oil are separated from each other without becoming a uniform layer of liquid in the refrigerant vessel under a use environment of the refrigerant and the refrigerating machine oil.

That the density of the refrigerating machine oil is higher than the density of the refrigerant means that the density of the refrigerating machine oil is higher than the density of the refrigerant in a liquid state under the use environment of the refrigerant and the refrigerating machine oil.

The height position of the upper end of the second refrigerant pipe is preferably equal to or lower than a position 10 mm higher than the lowermost end of the inner peripheral surface of the bottom part of the vessel body.

SUPPLEMENTARY NOTE

Although the embodiments of the present disclosure have been described above, it will be understood that various

36

changes in form and details can be made without departing from the gist and scope of the present disclosure described in the claims.

REFERENCE SIGNS LIST

- 1: refrigeration cycle apparatus
- 2: cascade unit
- 2x: cascade casing
- 3a: first utilization unit
- 3b: second utilization unit
- 3c: third utilization unit
- 5: primary-side unit
- 5a: primary-side refrigerant circuit (first refrigerant circuit)
- 10: secondary-side refrigerant circuit (refrigerant circuit, second refrigerant circuit)
- 12: cascade circuit
- 13a, 13b, 13c: utilization circuit
- 18: intermediate-pressure receiver (refrigerant vessel)
- 20: cascade-side control unit
- 21: secondary-side compressor (compressor)
- 21a: compressor motor
- 22: secondary-side switching mechanism
- 22a: first switching valve
- 22b: second switching valve
- 22x: discharge-side connection portion
- 22y: suction-side connection portion
- 23: suction flow path (refrigerant flow path)
- 24: discharge flow path
- 25: third pipe
- 26: fourth pipe
- 27: fifth pipe
- 28: first pipe
- 29: second pipe
- 30: secondary-side accumulator
- 34: oil separator
- 35: cascade heat exchanger
- 35a: secondary-side flow path
- 35b: primary-side flow path
- 36: cascade expansion valve
- 45: secondary-side receiver (refrigerant vessel)
- 46: bypass circuit (refrigerant flow path)
- 46a: bypass expansion valve
- 47: secondary-side subcooling heat exchanger
- 48: secondary-side subcooling circuit
- 48a: secondary-side subcooling expansion valve
- 50a-c: utilization-side control unit
- 51a-c: utilization-side expansion valve
- 52a-c: utilization-side heat exchanger
- 53a-c: indoor fan
- 60a, 60b, 60c: branch unit control unit
- 66a, 66b, 66c: first regulating valve
- 67a, 67b, 67c: second regulating valve
- 68a, 68b, 68c: check valve
- 69a, 69b, 69c: bypass pipe
- 70: primary-side control unit
- 71: primary-side compressor
- 72: primary-side switching mechanism
- 74: primary-side heat exchanger
- 76: primary-side first expansion valve
- 80: control unit
- 90: vessel body
- 91: upper part
- 92: lower peripheral surface portion
- 93: bottom part
- 93a: upper surface

37

93b: lower surface
 96: first refrigerant pipe
 97: second refrigerant pipe
 97a: upper end
 98: third refrigerant pipe
 102: primary-side second expansion valve
 103: primary-side subcooling heat exchanger
 104: primary-side subcooling circuit
 104a: primary-side subcooling expansion valve
 105: primary-side accumulator
 111: primary-side first connection pipe
 112: primary-side second connection pipe
 113: first refrigerant pipe
 114: second refrigerant pipe
 193a: upper curved surface (curved portion)
 193b: lower curved surface
 196: first refrigerant pipe
 197: second refrigerant pipe
 197a: upper end
 210: refrigerant circuit

CITATION LIST

Patent Literature

Patent Literature 1: JP 2008-164225 A

The invention claimed is:

1. A refrigerant vessel that reserves a refrigerant containing a carbon dioxide refrigerant in a refrigerant circuit in which the refrigerant and a refrigerating machine oil circulate, the refrigerant vessel comprising:

a vessel body including a cylindrical member and a bottom member that is welded and fixed to the cylindrical member and has a vertical direction as the thickness direction;

a first refrigerant pipe connected to the vessel body;

a second refrigerant pipe connected to the vessel body; and

a third refrigerant pipe connected to the vessel body, wherein

the bottom member has a concave curved surface facing a space inside the refrigerant vessel and is concaved downwardly,

the refrigerating machine oil is incompatible with the refrigerant and has a density higher than a density of the refrigerant,

an opening of the first refrigerant pipe is above an opening of the second refrigerant pipe and below an opening of the third refrigerant pipe in the vessel body,

the first refrigerant pipe includes a portion that extends in a downward direction within the vessel body and has an opening formed to be inclined with respect to an axial direction in which the first refrigerant pipe extends,

the second refrigerant pipe extends in the downward direction from the bottom member of the vessel body,

an opening of the second refrigerant pipe is offset from the opening of the first refrigerant pipe in a second direction perpendicular to the downward direction such that

when the refrigerating machine oil is inserted into the vessel body via the opening of the first refrigerant pipe, the refrigerating machine oil guided by the concave curved surface of the bottom member to the opening of the second refrigerant pipe, the opening of the first refrigerant pipe being inclined away from the opening of the second refrigerant pipe, and

the bottom member includes an indented cylindrical member with an indented cylindrical shape, with the

38

vertical direction as the thickness direction, and includes the downward concave curved surface facing a space inside the refrigerant vessel forming an indent in the indented cylindrical shape.

2. The refrigerant vessel according to claim 1, wherein a height position of an upper end of the second refrigerant pipe is equal to or lower than a lowermost end of an inner peripheral surface of the bottom member of the vessel body.

3. The refrigerant vessel according to claim 2, wherein the refrigerant vessel is a high-pressure receiver in which a high-pressure refrigerant is reserved, the high-pressure receiver being provided at a portion of the refrigerant circuit in which the high-pressure refrigerant flows.

4. The refrigerant vessel according to claim 2, wherein the refrigerant circuit includes a portion in which a high-pressure refrigerant, a low-pressure refrigerant, and an intermediate-pressure refrigerant flow, and the refrigerant vessel is an intermediate-pressure receiver in which the intermediate-pressure refrigerant is reserved, the intermediate-pressure receiver being provided at a portion of the refrigerant circuit in which the intermediate-pressure refrigerant flows.

5. The refrigerant vessel according to claim 2, wherein the refrigerant circuit includes a compressor, and the refrigerant vessel further includes a third refrigerant pipe that is connected to the vessel body and guides the refrigerant in the vessel body to a suction side of the compressor.

6. The refrigerant vessel according to claim 2, wherein the refrigerating machine oil includes polyalkylene glycol oil.

7. A refrigeration cycle apparatus comprising the refrigerant circuit including the refrigerant vessel according to claim 2.

8. The refrigerant vessel according to claim 1, wherein the refrigerant vessel is a high-pressure receiver in which a high-pressure refrigerant is reserved, the high-pressure receiver being provided at a portion of the refrigerant circuit in which the high-pressure refrigerant flows.

9. The refrigerant vessel according to claim 8, wherein the refrigerant circuit includes a compressor, and the refrigerant vessel further includes a third refrigerant pipe that is connected to the vessel body and guides the refrigerant in the vessel body to a suction side of the compressor.

10. The refrigerant vessel according to claim 8, wherein the refrigerating machine oil includes polyalkylene glycol oil.

11. A refrigeration cycle apparatus comprising the refrigerant circuit including the refrigerant vessel according to claim 8.

12. The refrigerant vessel according to claim 1, wherein the refrigerant circuit includes a portion in which a high-pressure refrigerant, a low-pressure refrigerant, and an intermediate-pressure refrigerant flow, and the refrigerant vessel is an intermediate-pressure receiver in which the intermediate-pressure refrigerant is reserved, the intermediate-pressure receiver being provided at a portion of the refrigerant circuit in which the intermediate-pressure refrigerant flows.

13. The refrigerant vessel according to claim 12, wherein the refrigerant circuit includes a compressor, and the refrigerant vessel further includes a third refrigerant pipe that is connected to the vessel body and guides the refrigerant in the vessel body to a suction side of the compressor.

39

14. The refrigerant vessel according to claim 12, wherein the refrigerating machine oil includes polyalkylene glycol oil.

15. The refrigerant vessel according to claim 1, wherein the refrigerant circuit includes a compressor, and the third refrigerant pipe is connected to the vessel body and guides the refrigerant in the vessel body to a suction side of the compressor.

16. The refrigerant vessel according to claim 15, wherein the refrigerating machine oil includes polyalkylene glycol oil.

17. The refrigerant vessel according to claim 1, wherein the refrigerating machine oil includes polyalkylene glycol oil.

18. A refrigeration cycle apparatus comprising the refrigerant circuit including the refrigerant vessel according to claim 1.

19. The refrigeration cycle apparatus according to claim 18, comprising:

a first refrigerant circuit in which a refrigerant flows; and a cascade heat exchanger,

wherein the refrigerant circuit is provided as a second refrigerant circuit independent of the first refrigerant circuit,

the cascade heat exchanger exchanges heat between the refrigerant flowing in the first refrigerant circuit and the refrigerant flowing in the second refrigerant circuit, and the refrigerant in the second refrigerant circuit that has become a liquid refrigerant by being cooled by the refrigerant in the first refrigerant circuit in the cascade heat exchanger is reserved in the refrigerant vessel.

20. A refrigeration cycle apparatus comprising:
a first refrigerant circuit in which a first refrigerant flows;
and

40

a second refrigerant circuit in which a refrigerating machine oil and a second refrigerant containing a carbon dioxide refrigerant circulate, the second refrigerant circuit is independent of the first refrigerant circuit; and

a cascade heat exchanger exchanges heat between the first refrigerant flowing in the first refrigerant circuit and the second refrigerant flowing in the second refrigerant circuit,

wherein the first refrigerant circuit has a compressor sucking the first refrigerant passed through the cascade heat exchanger,

the second refrigerant circuit has a refrigerant vessel that reserves the second refrigerant and an expansion valve provided between the cascade heat exchanger and the refrigerant vessel,

the refrigerant vessel has a vessel body, a first refrigerant pipe connected to the vessel body, and a second refrigerant pipe connected to the vessel body,

the refrigerant machine oil is incompatible with the second refrigerant and has a density higher than a density of the second refrigerant,

the second refrigerant pipe extends downward from a bottom of the vessel body,

the second refrigerant that has become a liquid refrigerant by being cooled by the first refrigerant in the cascade heat exchanger is reserved in the refrigerant vessel, and

the compressor or both the compressor and the expansion valve are controlled so that the second refrigerant flowing from the cascade heat exchanger to the refrigerant vessel does not exceed a critical point.

* * * * *